INCH-POUND

MIL-M-28787/505
23 JUNE 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

IOC DATA BUS LOWER MODULE, KEY CODE AWG

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Input/Output Controller (IOC) data bus lower module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.405 failures per million operating hours for class I or 3.215 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AWG.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	----------Module shall pass all remaining tests--------						
D1	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D2,D3	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D4,D5	Ibd	V	S	-0.12	1.11	-0.12	1.11	-0.12	1.11	mA
D6	Ibd	V	S	-0.12	2.11	-0.12	2.11	-0.12	2.11	mA
D7-D12	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D13	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D14-D16	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D17	Ibd	V	S	0.0	11.2	0.0	11.2	0.0	11.2	mA
D18	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D19	Ibd	V	S	0.0	2.9	0.0	2.9	0.0	2.9	mA
D20	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D21-D25	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D26	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D27,D28	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D29	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D30	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D31	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D32-D35	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D36,D37	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D38	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
E1	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E2,E3	Ios	VI	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
E4-E7	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E8	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E9-E12	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E13	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E14-E18	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E19-E26	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E27-E30	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E31-E34	Ios	VI	S	-100.0	-20.0	-100.0	-20.0	-100.0	-20.0	mA
E35-E38	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E39	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E40	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E41-E49	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E50,E51	Ios	VI	S	-155.0	-65.8	-155.0	-65.8	-155.0	-65.8	mA
E52	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1	Iih	VII	X	-2.25	0.07	-2.25	0.07	-2.25	0.07	mA
F2,F3	Iih	VII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
F4,F5	Iih	VII	X	-5.68	-2.85	-5.68	-2.85	-5.68	-2.85	mA
F6	Iih	VII	X	-7.68	-2.8	-7.68	-2.8	-7.68	-2.8	mA
F7-F12	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F13	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F14-F16	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F17	Iih	VII	X	-21.4	0.57	-21.4	0.57	-21.4	0.57	mA
F18	Iih	VII	X	-2.8	0.09	-2.8	0.09	-2.8	0.09	mA
F19	Iih	VII	X	-8.45	0.26	-8.45	0.26	-8.45	0.26	mA
F20	Iih	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F21-F23	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F24-F27	Iih	VII	X	-0.45	0.09	-0.45	0.09	-0.45	0.09	mA
F28,F29	Iih	VII	X	-1.8	0.02	-1.8	0.02	-1.8	0.02	mA
F30	Iih	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F31	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F32-F39	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F40	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F41	Iih	VII	X	-1.4	0.06	-1.4	0.06	-1.4	0.06	mA

- See footnotes at end of table.

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TABLE I. *Module electrical requirements* – Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5 °C 85 ± 5 C		Units
				Min	Max	Min	Max	Min	Max	
F42	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F43	Iih	VII	X	-1.2	0.06	-1.2	0.06	-1.2	0.06	mA
F44-F47	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F48,F49	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F50-F53	Iih	VII	X	-0.45	0.09	-0.45	0.09	-0.45	0.09	mA
F54	Iih	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F55,F56	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F57,F58	Iih	VII	X	-0.85	0.09	-0.85	0.09	-0.85	0.09	mA
F59	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
F60,F61	Iih	VII	X	-5.68	-2.8	-5.68	-2.8	-5.68	-2.8	mA
F62	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G1	Iil	VIII	X	-2.25	0.07	-2.25	0.07	-2.25	0.07	mA
G2,G3	Iil	VIII	X	-0.25	0.03	-0.25	0.03	-0.25	0.03	mA
G4,G5	Iil	VIII	X	-5.68	-2.85	-5.68	-2.85	-5.68	-2.85	mA
G6	Iil	VIII	X	-7.68	-2.8	-7.68	-2.8	-7.68	-2.8	mA
G7-G12	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G13	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G14-G16	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G17	Iil	VIII	X	-21.4	0.57	-21.4	0.57	-21.4	0.57	mA
G18	Iil	VIII	X	-2.8	0.09	-2.8	0.09	-2.8	0.09	mA
G19	Iil	VIII	X	-8.45	0.26	-8.45	0.26	-8.45	0.26	mA
G20	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G21-G23	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G24-G27	Iil	VIII	X	-0.45	0.09	-0.45	0.09	-0.45	0.09	mA
G28,G29	Iil	VIII	X	-1.8	0.02	-1.8	0.02	-1.8	0.02	mA
G30	Iil	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G31	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G32-G39	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G40	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G41	Iil	VIII	X	-1.4	0.06	-1.4	0.06	-1.4	0.06	mA
G42	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA

– See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ∓ 5°C		Class II 4/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
G43	Iil	VIII	X	-1.2	0.06	-1.2	0.06	-1.2	0.06	mA
G44-G47	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G48,G49	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G50-G53	Iil	VIII	X	-0.45	0.09	-0.45	0.09	-0.45	0.09	mA
G54	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G55,G56	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G57,G58	Iil	VIII	X	-0.85	0.09	-0.85	0.09	-0.85	0.09	mA
G59	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G60,G61	Iil	VIII	X	-5.68	-2.8	-5.68	-2.8	-5.68	-2.8	mA
G62	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
H1-H52	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J52	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L2,L3	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L4-L24	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L25	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L26	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L27	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L28-L34	Vic	XI	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L35	Vic	XI	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
M1	PSFC	XII	S	4.2	7.8	4.2	7.8	4.2	7.8	uF
M2	PSFC	XII	S	4.2	7.8	4.2	7.8	4.2	7.8	uF
N	Icc	XIII	X	415	998	415	998	415	998	mA
P1,P2	Functional	XIV	X	See table XIV						
Q1,Q2	Dynamic	XV	S	See table XV						
Q3,Q4	Dynamic	XV	Q	See table XV						

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R1	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R2.R3	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R4	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R5	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R6	Ci/o	XVII	Q	6	125	6	125	6	125	pF
R7-R15	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R16	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R17	Ci/o	XVII	Q	20	211	20	211	20	211	pF
R18	Ci/o	XVII	Q	6	100	6	100	6	100	pF
R19	Ci/o	XVII	Q	16	225	16	225	16	225	pF
R20	Ci/o	XVII	Q	4	125	4	125	4	125	pF
R21-R25	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R26	Ci/o	XVII	Q	4	125	4	125	4	125	pF
R27,R28	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R29	Ci/o	XVII	Q	6	100	6	100	6	100	pF
R30	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R31	Ci/o	XVII	Q	6	100	6	100	6	100	pF
R32-R35	Ci/o	XVII	Q	2	50	2	50	2	50	pF
R36	Ci/o	XVII	Q	4	100	4	100	4	100	pF
R37	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R38-R42	Ci/o	XVII	Q	2	75	2	75	2	75	pF
R43-R50	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R51-R53	Ci/o	XVII	Q	6	100	6	100	6	100	pF
R54	Ci/o	XVII	Q	6	75	6	75	6	75	pF
R55,R56	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R57	Ci/o	XVII	Q	6	100	6	100	6	100	pF
R58	Ci/o	XVII	Q	6	125	6	125	6	125	pF
R59	Ci/o	XVII	Q	4	75	4	75	4	75	pF
R60,R61	Ci/o	XVII	Q	6	100	6	100	6	100	pF
R62	Ci/o	XVII	Q	4	75	4	75	4	75	pF
S	Input diode latchup test	XVIII	Q	-1.2	2.4	-1.2	2.4	-1.2	2.4	V
T1,T2	PSTA	XIX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/505-1 for class I and M28787/505-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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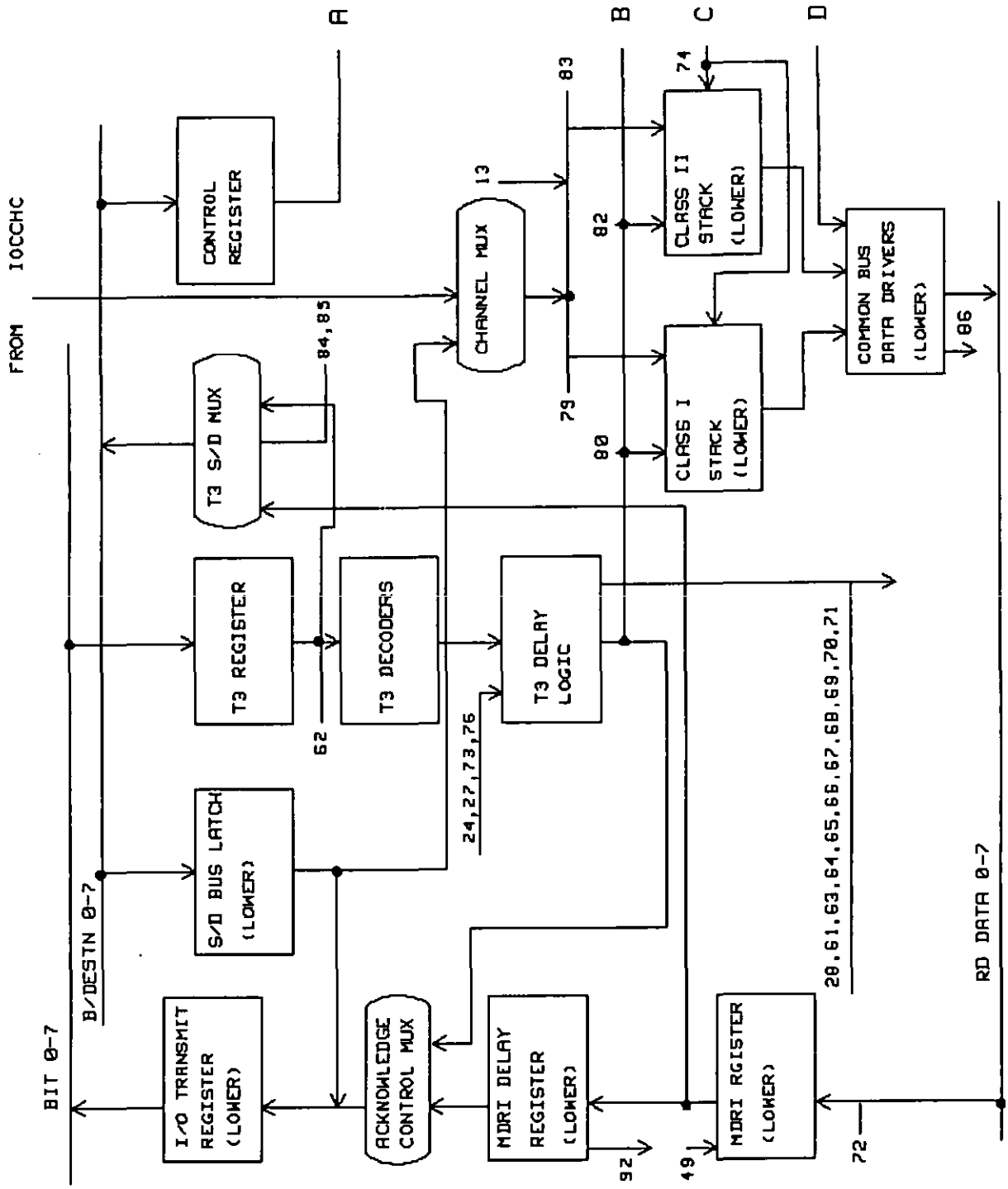


FIGURE 1. AWG functional block diagram.

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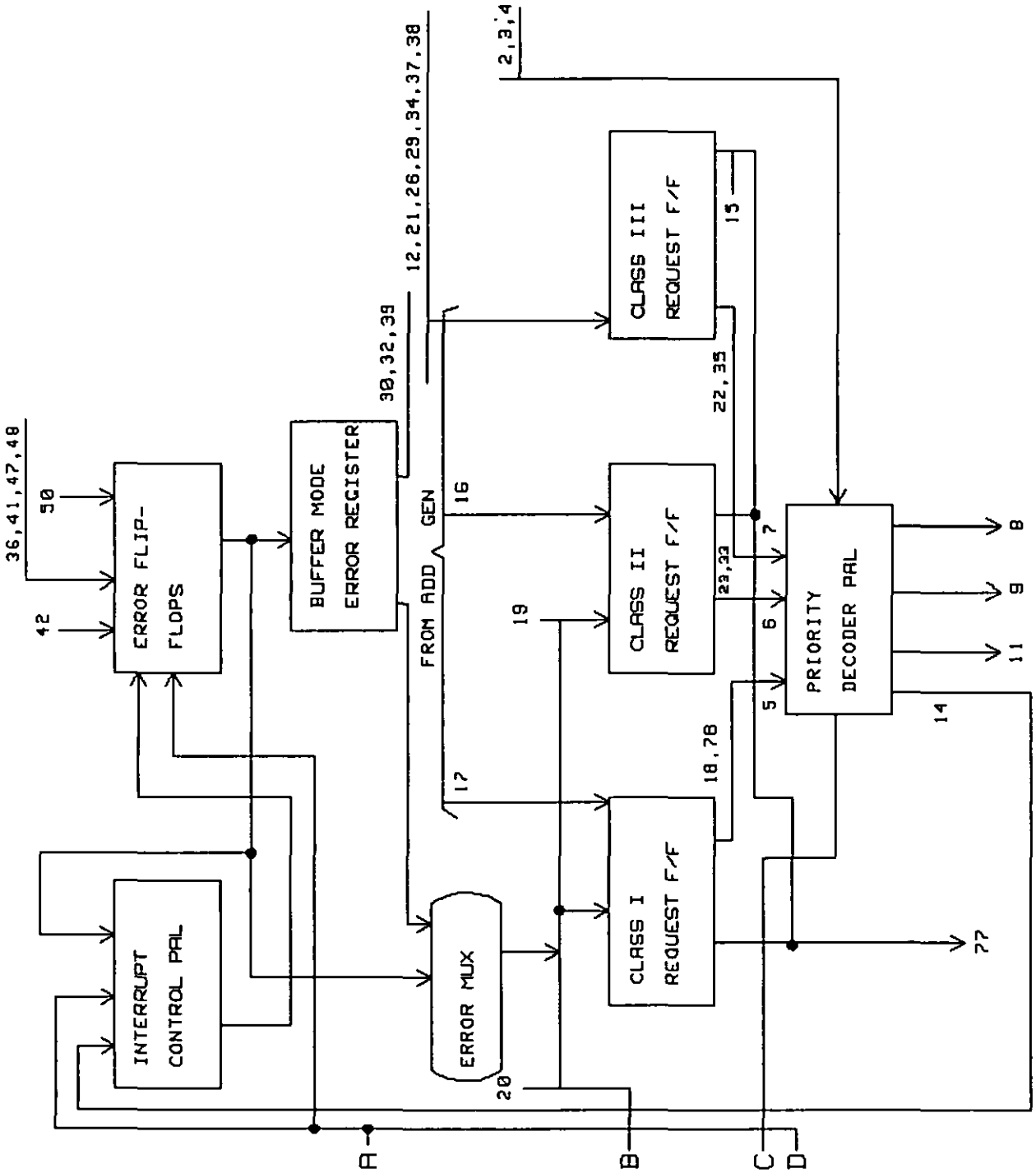


FIGURE 1. AWG functional block diagram - Continued.

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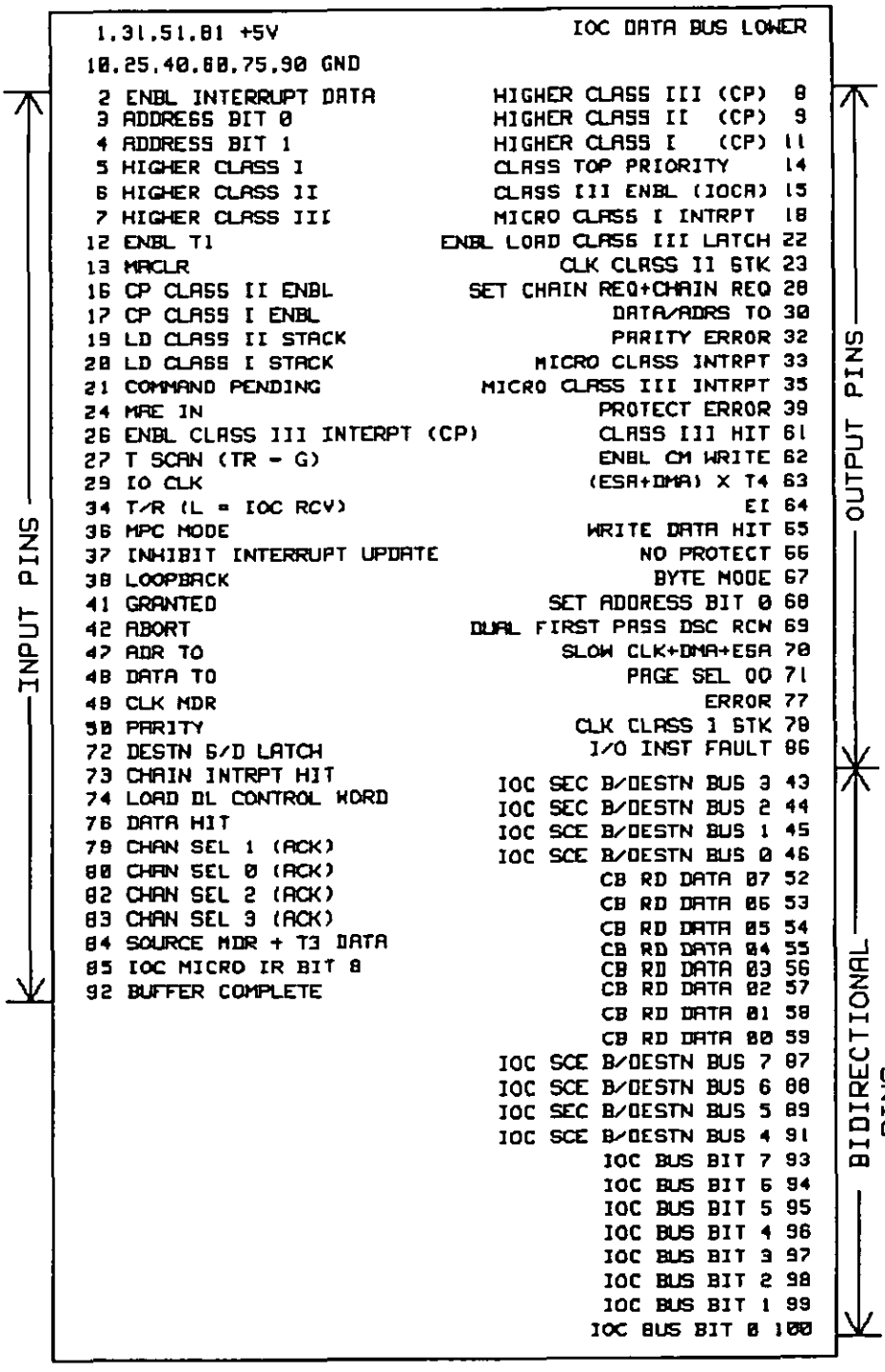


FIGURE 2. AWG pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, L, and S are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, L or S are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, L, and S where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

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4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

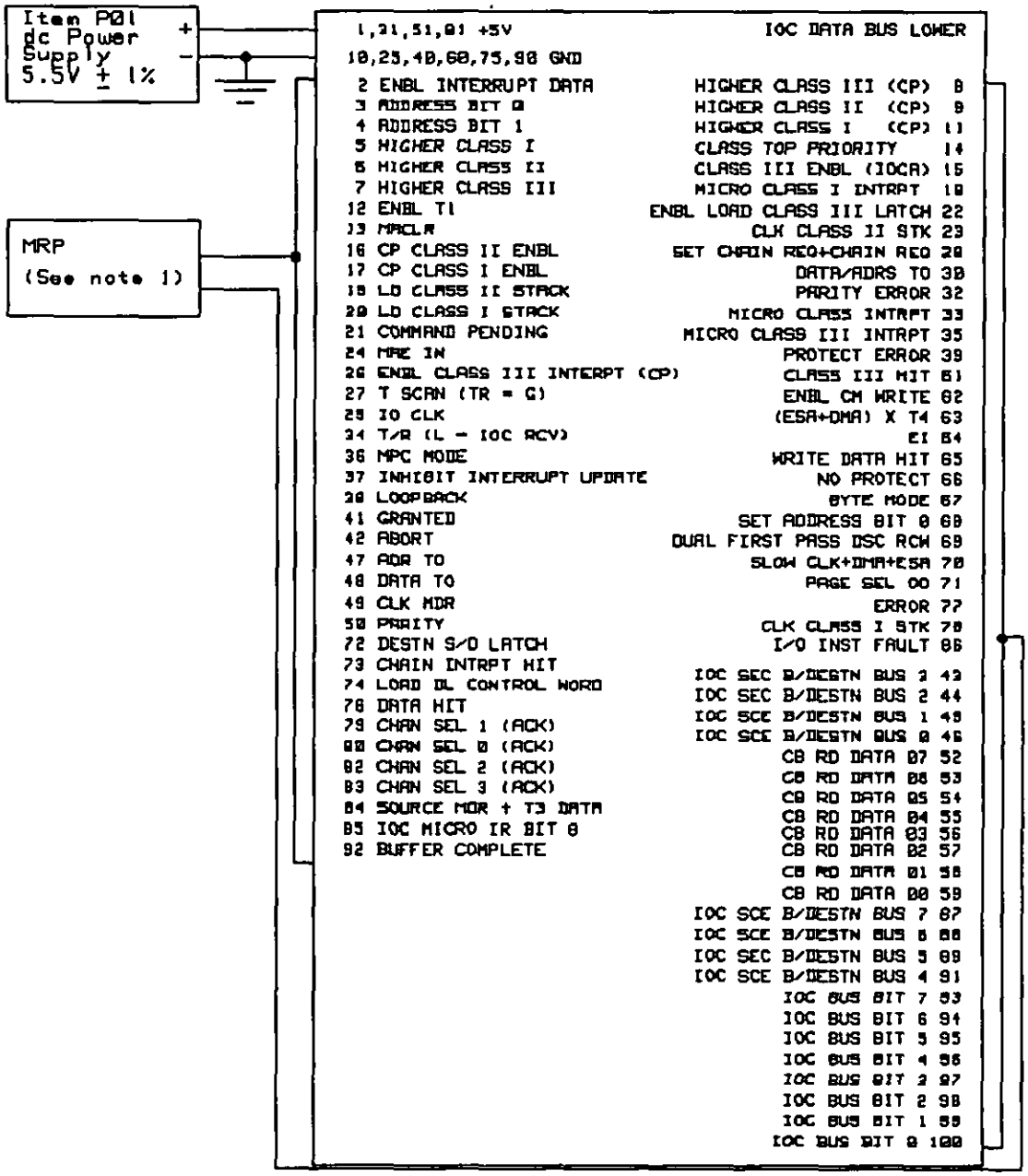
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through J and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. AWG operating temperature and component temperature test circuit.

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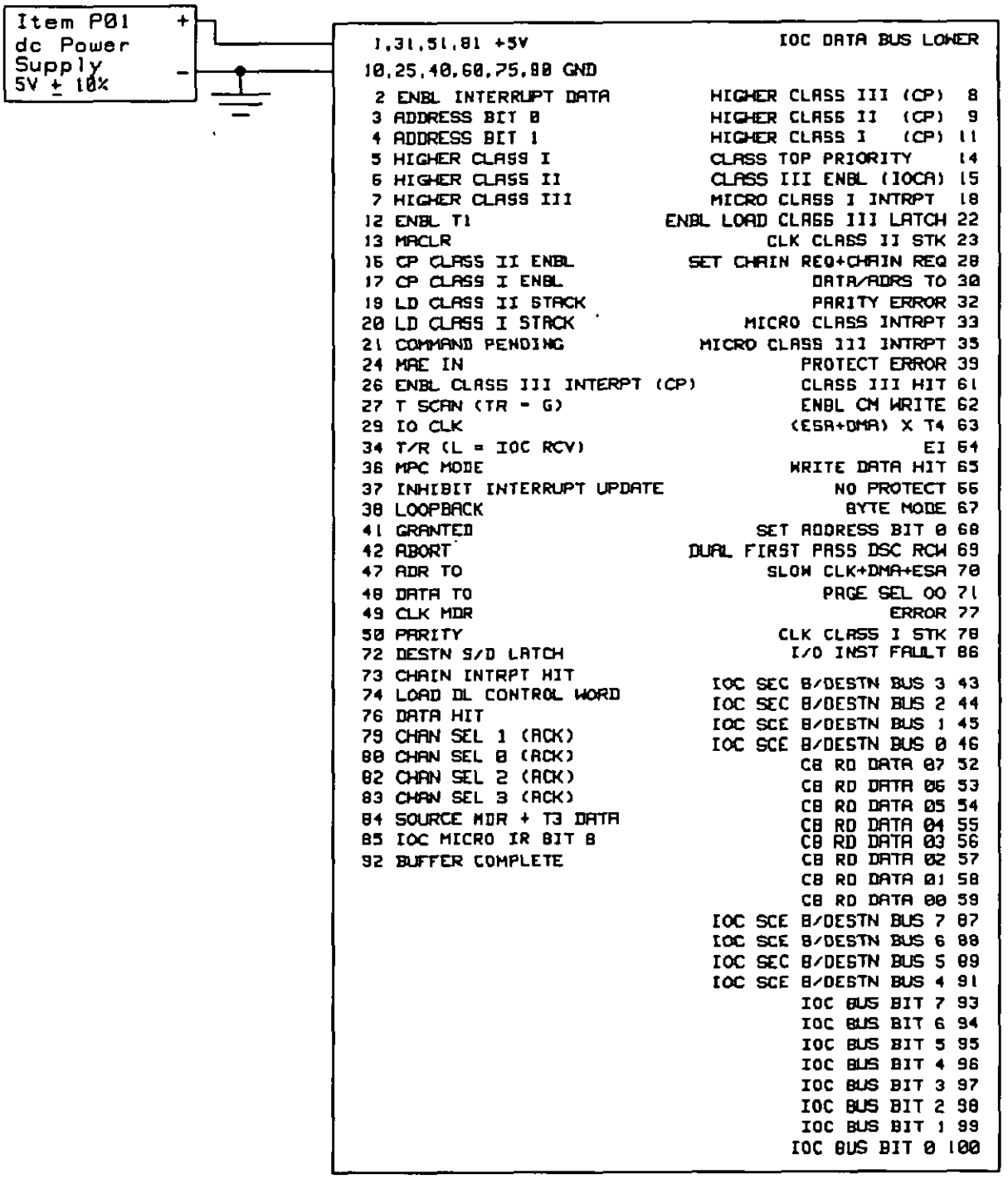
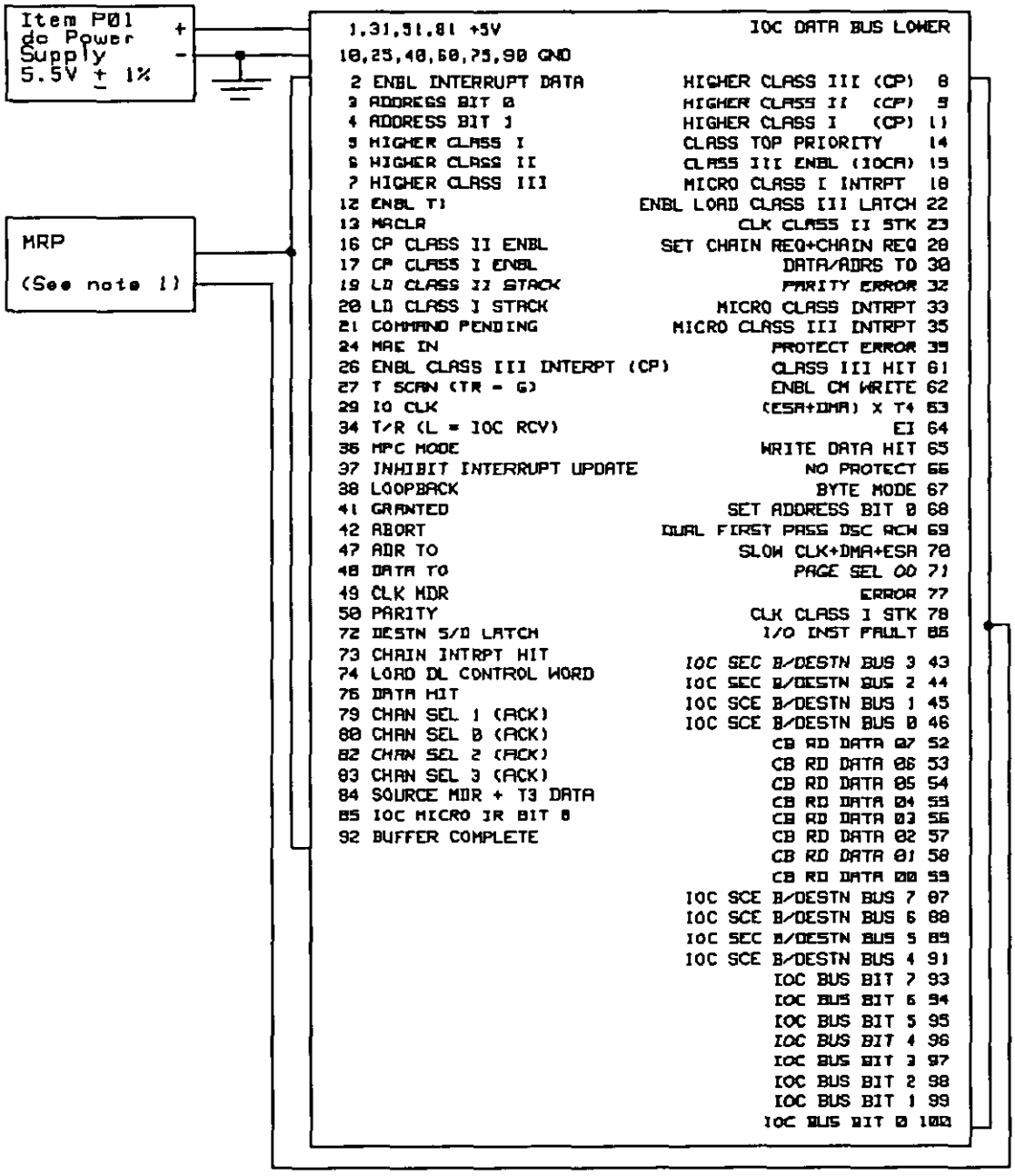


FIGURE 4. AWG vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. AWG life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) NA = not applicable.
(m) PSFC = power supply filter capacitance.
(n) PSTA = power supply transient amplitude.
(o) Vic = input clamp diode voltage.
(p) Vih = input high level voltage.
(q) Vil = input low level voltage.
(r) Vin = input forcing voltage.
(s) Voh = output high level voltage.
(t) Vol = output low level voltage.
(u) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 5.478 W (max); 3.673 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
EN INTR DATA	-1.2	0.4	2.4	-2.25	0.07	1.2	75
ADRS 0	-1.5	0.4	2.4	-0.25	0.03	1.0	50
ADRS 1	-1.5	0.4	2.4	-0.25	0.03	1.0	50
CI	NA	0.4	2.4	-5.68	-2.85	1.11	75
CII	NA	0.4	2.4	-5.68	-2.85	1.11	100
CIII	NA	0.4	2.4	-7.68	-2.8	2.11	125
ENBL T1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MACLR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CII EN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CI EN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
LD CII STK	-1.2	0.4	2.4	-0.6	0.02	0.1	75
LD CI STK	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COMM PNDNG	-1.2	0.4	2.4	-2.0	0.05	1.0	75
MAE IN	-1.2	0.4	2.4	-0.6	0.02	0.1	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
EN CIII INTR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
T SCAN	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IO CLK	-1.2	0.4	2.4	-21.4	0.57	11.2	211
T/R	-1.2	0.4	2.4	-2.8	0.09	1.2	100
MPC MODE	-1.2	0.4	2.4	-8.45	0.26	2.9	225
IN INTR UP	-1.2	0.4	2.4	-0.8	0.04	0.2	125
LOOPBACK	-1.2	0.4	2.4	-0.6	0.02	0.1	50
GRANTED	-1.2	0.4	2.4	-0.6	0.02	0.1	50
ABORT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
B/DESTN 3	NA	0.4	2.4	-0.45	0.09	NA	75
B/DESTN 2	NA	0.4	2.4	-0.45	0.09	NA	75
B/DESTN 1	NA	0.4	2.4	-0.45	0.09	NA	75
B/DESTN 0	NA	0.4	2.4	-0.45	0.09	NA	75
ADRS TO	-1.2	0.4	2.4	-1.8	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DATA TO	-1.2	0.4	2.4	-1.8	0.02	0.1	50
CLK MDR	-1.2	0.4	2.4	-0.8	0.04	0.2	125
PARITY	-1.2	0.4	2.4	-0.6	0.02	0.1	50
RD DATA 07	NA	0.4	2.4	-0.25	0.07	NA	75
RD DATA 06	NA	0.4	2.4	-0.25	0.07	NA	75
RD DATA 05	NA	0.4	2.4	-0.25	0.07	NA	75
RD DATA 04	NA	0.4	2.4	-0.25	0.07	NA	75
RD DATA 03	NA	0.4	2.4	-0.25	0.07	NA	75
RD DATA 02	NA	0.4	2.4	-0.25	0.07	NA	75
RD DATA 01	NA	0.4	2.4	-0.25	0.07	NA	75
RD DATA 00	NA	0.4	2.4	-0.25	0.07	NA	75
S/D LTC	-1.5	0.4	2.4	-0.2	0.02	0.1	50
CH INTR HIT	-1.2	0.4	2.4	-1.4	0.06	0.3	100
LD CTRL WD	-1.5	0.4	2.4	-0.2	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
DATA HIT	-1.2	0.4	2.4	-1.2	0.06	0.3	100
CHAN 1	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CHAN 0	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CHAN 2	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CHAN 3	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MDR + T3	-1.2	0.4	2.4	-1.2	0.04	0.2	100
IR BIT 8	-1.2	0.4	2.4	-1.2	0.04	0.2	75
B/DESTN 7	NA	0.4	2.4	-0.45	0.09	NA	100
B/DESTN 6	NA	0.4	2.4	-0.45	0.09	NA	100
B/DESTN 5	NA	0.4	2.4	-0.45	0.09	NA	100
B/DESTN 4	NA	0.4	2.4	-0.45	0.09	NA	75
BUFFER COMP	-1.5	0.4	2.4	-0.2	0.02	0.1	75
BIT 7	-1.5	0.4	2.4	-0.25	0.07	NA	75
BIT 6	-1.5	0.4	2.4	-0.25	0.07	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
BIT 5	-1.5	0.4	2.4	-0.85	0.09	NA	100
BIT 4	-1.5	0.4	2.4	-0.85	0.09	NA	125
BIT 3	-1.5	0.4	2.4	-0.25	0.07	NA	75
BIT 2	-1.5	0.4	2.4	-5.68	-2.8	NA	100
BIT 1	-1.5	0.4	2.4	-5.68	-2.8	NA	100
BIT 0	-1.5	0.4	2.4	-0.25	0.07	NA	75

6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
CIII (CP)	2.4	0.5	-0.4	6.0	-40.0	NA
CII (CP)	2.4	0.5	-0.16	1.6	-30.0	NA
CI (CP)	2.4	0.5	-0.16	1.6	-30.0	NA
TOP PRIORITY	2.4	0.5	-0.4	6.0	-60.0	NA
CIII EN	2.4	0.5	-0.4	6.0	-60.0	NA
CI INTR	2.4	0.5	-0.4	6.0	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
EN CIII LTC	2.4	0.5	-0.4	6.0	-60.0	NA
CLK CII STK	2.4	0.5	-0.4	6.0	-40.0	NA
SET CH REQ	2.4	0.5	-0.4	6.0	-60.0	NA
DATA/ADRS TO	2.4	0.5	-0.4	6.0	-60.0	NA
PARITY ERR	2.4	0.5	-0.4	6.0	-60.0	NA
CLASS INTR	2.4	0.5	-0.4	6.0	-60.0	NA
CIII INTR	2.4	0.5	-0.16	1.6	-30.0	NA
PROTECT ERR	2.4	0.5	-0.4	6.0	-60.0	NA
B/DESTN 3	2.4	0.5	-0.4	6.0	-60.0	75
B/DESTN 2	2.4	0.5	-0.4	6.0	-60.0	75
B/DESTN 1	2.4	0.5	-0.4	6.0	-60.0	75
B/DESTN 0	2.4	0.5	-0.4	6.0	-60.0	75
RD DATA 07	2.4	0.5	-0.4	6.0	-100.0	75
RD DATA 06	2.4	0.5	-0.4	6.0	-100.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
RD DATA 05	2.4	0.5	-0.4	6.0	-100.0	75
RD DATA 04	2.4	0.5	-0.4	6.0	-100.0	75
RD DATA 03	2.4	0.5	-0.4	6.0	-100.0	75
RD DATA 02	2.4	0.5	-0.4	6.0	-100.0	75
RD DATA 01	2.4	0.5	-0.4	6.0	-100.0	75
RD DATA 00	2.4	0.5	-0.4	6.0	-100.0	75
CIII HIT	2.4	0.5	-0.4	6.0	-60.0	NA
EN CM WR	2.4	0.5	-0.4	6.0	-60.0	NA
T4	2.4	0.5	-0.4	6.0	-60.0	NA
EI	2.4	0.5	-0.4	6.0	-60.0	NA
WR DATA HIT	2.4	0.5	-0.16	1.6	-20.0	NA
NO PROTECT	2.4	0.5	-0.16	1.6	-20.0	NA
BYTE MODE	2.4	0.5	-0.16	1.6	-20.0	NA
SET ADRS 0	2.4	0.5	-0.16	1.6	-20.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
DUAL1ST PASS	2.4	0.5	-0.4	6.0	-60.0	NA
SLOW	2.4	0.5	-0.4	6.0	-60.0	NA
PAGE SEL 00	2.4	0.5	-0.4	6.0	-60.0	NA
ERROR	2.4	0.5	-0.4	6.0	-60.0	NA
CLK CI STK	2.4	0.5	-0.4	6.0	-40.0	NA
FAULT	2.4	0.5	-0.16	1.6	-30.0	NA
B/DESTN 7	2.4	0.5	-0.4	6.0	-60.0	100
B/DESTN 6	2.4	0.5	-0.4	6.0	-60.0	100
B/DESTN 5	2.4	0.5	-0.4	6.0	-60.0	100
B/DESTN 4	2.4	0.5	-0.4	6.0	-60.0	75
BIT 7	2.4	0.5	-0.4	6.0	-60.0	75
BIT 6	2.4	0.5	-0.4	6.0	-60.0	75
BIT 5	2.4	0.5	-0.4	6.0	-60.0	100
BIT 4	2.4	0.5	-0.4	6.0	-60.0	125

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
BIT 3	2.4	0.5	-0.4	6.0	-60.0	75
BIT 2	2.4	0.5	-0.4	6.0	-65.8	100
BIT 1	2.4	0.5	-0.4	6.0	-65.8	100
BIT 0	2.4	0.5	-0.4	6.0	-60.0	75

6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MPC MODE to PROTECT ERR	-	-	-	23	23	23	nS	12
2	tpd	MPC MODE to PARITY ERR	-	-	-	33	33	33	nS	12
2	tpd	MPC MODE to DATA/ ADRS TO	-	-	-	33	33	33	nS	12
3	tpd	MPC MODE to ERROR	-	-	-	59	59	59	nS	12
4	tpd	IO CLK to EN CIII LTC	-	-	-	55	55	55	nS	12
5	tpd	IO CLK to CIII INTR	-	-	-	36	36	36	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
6	tpd	IO CLK to CIII EN	-	-	-	29	29	29	nS	12
7	tpd	IO CLK to CLK CII STK	-	-	-	17	17	17	nS	12
8	tpd	IO CLK to CLK CI STK	-	-	-	17	17	17	nS	12
9	tpd	IO CLK to CI INTR	-	-	-	45	45	45	nS	12
9	tpd	IO CLK to CLASS INTR	-	-	-	45	45	45	nS	12
10	tpd	IO CLK to CIII HIT	-	-	-	39	39	39	nS	12
11	tpd	IO CLK to SET CH REQ	-	-	-	30	30	30	nS	12
12	tpd	CI to CI (CP)	-	-	-	52	52	52	nS	12
12	tpd	CII to CI (CP)	-	-	-	52	52	52	nS	12
12	tpd	ADRS 0 to CI (CP)	-	-	-	52	52	52	nS	12
12	tpd	ADRS 1 to CI (CP)	-	-	-	52	52	52	nS	12
13	tpd	CI to TOP PRIORITY	-	-	-	59	59	59	nS	12
13	tpd	CII to TOP PRI-ORITY	-	-	-	59	59	59	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
13	tpd	CIII to TOP PRI-ORITY	-	-	-	59	59	59	nS	12
13	tpd	ADRS 0 to TOP PRI-ORITY	-	-	-	59	59	59	nS	12
13	tpd	ADRS 1 to TOP PRI-ORITY	-	-	-	59	59	59	nS	12
14	tpd	CIII to CIII (CP)	-	-	-	17	17	17	nS	12
15	tpd	IO CLK to DUAL1ST PASS	-	-	-	32	32	32	nS	12
15	tpd	IO CLK to EI	-	-	-	32	32	32	nS	12
16	tpd	IO CLK to SET ADRS 0	-	-	-	58	58	58	nS	12
17	tpd	IO CLK to SLOW	-	-	-	63	63	63	nS	12
18	tpd	IO CLK to T4	-	-	-	44	44	44	nS	12
19	tpd	IO CLK to WR DATA HIT	-	-	-	58	58	58	nS	12
19	tpd	IO CLK to NO PRO-TECT	-	-	-	58	58	58	nS	12
19	tpd	IO CLK to BYTE MODE	-	-	-	58	58	58	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
20	tpd	IO CLK to PAGE SEL 00	-	-	-	70	70	70	nS	12
21	tpd	IO CLK to EN CM WR	-	-	-	51	51	51	nS	12
22	tpd	IO CLK to B/DESTN 3	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 2	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 1	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 0	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 7	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 6	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 5	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 4	-	-	-	147	147	147	nS	12
23	tpd	IO CLK to FAULT	-	-	-	36	36	36	nS	12
24	tpd	IO CLK to BIT 7	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 6	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 5	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 4	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 3	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 2	-	-	-	32	32	32	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
24	tpd	IO CLK to BIT 1	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 0	-	-	-	32	32	32	nS	12
25	tsu	ENBL T1 to IO CLK	26	26	26	-	-	-	nS	12
25	tsu	T/R to IO CLK	26	26	26	-	-	-	nS	12
25	tsu	LOOPBACK to IO CLK	26	26	26	-	-	-	nS	12
25	tsu	IN INTR UP to IO CLK	26	26	26	-	-	-	nS	12
26	tsu	DATA HIT to IO CLK	12	12	12	-	-	-	nS	12
27	tsu	CH INTR HIT to IO CLK	34	34	34	-	-	-	nS	12
28	tsu	RD DATA 07 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 06 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 05 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 04 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 03 to CLK MDR	10	10	10	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
28	tsu	RD DATA 02 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 01 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 00 to CLK MDR	10	10	10	-	-	-	nS	12

6.4 Module function. The IOC data bus interface is contained on two modules (IOC data bus upper (EWC) and IOC data bus lower (AWG)). The IOC data bus upper and IOC data bus lower modules function to interface the common bus to the IOC bus. The IOC data bus interface modules contain transmit registers, Memory Data Register Input (MDRI), the class I, II, and III interrupt stacks, Memory Address Register (MAR), Memory Data Register (MDR), error capture circuits, and interrupt request flip-flops. The transmit registers are used to transmit data and commands to the IOCs via the IOC bus. Data or commands that are destined for the Input/Output Channel Adapters (IOCAs) will first be placed into the MDRI registers. The class I, II, and III interrupt stacks hold interrupt data until sampled by the Central Processor (CP) at its convenience. The class I stack holds parity error and data/address timeout interrupts; the class II stack holds I/O instruction fault and protect interrupt data; and the class III stack holds channel data along with Intercomputer Timeout (ICTO), External Interrupt (EI), and input or output chain interrupts. IOC data bus upper and lower modules contain bits 15-18 and bits 7-0 of the interrupt word, respec- tively. The 16-bit MAR and MDR, both contained on the IOC data bus upper module, are used to hold the address and data destined for main memory during Externally Specified Addressing (ESA) mode and for writes. These registers will only be used during the buffer mode. Error capture circuits receive abort, timeout, and parity error information which is used to clock the class I, II, and III interrupt flip-flops. Error information is also placed onto the IOC bus for transfer to the IOCAs. The interrupt flip-flop outputs are used to send interrupt requests to the CP emulation control 2 module. IOC data bus lower contains the T3 register,

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interrupt stacks and control circuits, the transmit register, error capture circuits, the MDR, and associated logic and control circuits.

6.4.1 T3 register. The T3 register is an 8-bit register which receives IOC bus bits 7-0 as its inputs. These bits are encoded and represent a chain or interrupt request, a data buffer request, or an IOCA's response to an IOC command. The selected IOCA will place its request/response on the IOC bus data lines during T3 time. Outputs from the T3 register are sent through a decoding network and onto the T3 source/destination multiplexer for application to the source/destination bus and the T3 buffer control circuits.

6.4.2 Transmit register. The transmit register is an 8-bit register receiving inputs from the source B/destination bus and sending its outputs to the IOC bus bits 7-0.

6.4.3 Control register. The control register is an 8-bit register that receives source B/destination bus bits 7-0 as inputs. The outputs from the control register are applied to the interrupt control Programmable Array Logic (PAL).

6.4.4 Interrupt stacks. Inputs to the interrupt stack are via the channel select acknowledge multiplexer which selects either IO data out (bits 7-4) or channel select 3-0 signals from the IOC channel control module. Both class I and class II interrupt stacks are controlled by the stack control PAL. Common bus address bits 0 and 1 and an enable interrupt data signal from the IOC common bus control module along with the request flip-flop outputs are provided as inputs to the PAL. Outputs from the interrupt stacks are placed on the common bus via data driver circuitry. The common bus driver circuitry is enabled by the enable interrupt data signal from the IOC common bus control module.

6.4.5 Interrupt control and request control. The interrupt control PAL receives enables and clear signals for class I and class II interrupts along with data parity information and T1. All outputs from the interrupt control PAL are sent to error capture control. Request control circuits for class I and class II interrupts include the class I and class II interrupt flip-flops and synchronous gating circuits used in clocking the class I and class II interrupt stacks. The ANDing of class I and class II interrupts with a central processor class I and class II enable signal allows lockout of these interrupts by the CP, a class I interrupt will be lost if locked out; and a class II interrupt will not be locked out within the IOC but

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interrupt action will be prohibited if locked out by the CP. The input to the class III flip-flop is a class III "Hit" (chain request or external interrupt) gated with the appropriate enables at T1 time, and clocked by the IO clock. This class III request will be latched and applied to the data lower card priority decode PAL.

6.4.6 Memory data register. The MDR is an 8-bit register that receives common bus data bits 7-0 as inputs. The MDR is clocked by the clock MDR signal originating at the IOC common bus control module.

6.4.7 Description of I/O pins.

Signal name	Pin number	I/O	Description
EN INTR DATA	2	I	ENBL INTERRUPT DATA
ADRS 0	3	I	ADDRESS BIT 0
ADRS 1	4	I	ADDRESS BIT 1
CI	5	I	HIGHER CLASS I
CII	6	I	HIGHER CLASS II
CIII	7	I	HIGHER CLASS III
CIII (CP)	8	O	HIGHER CLASS III (CP)
CII (CP)	9	O	HIGHER CLASS II (CP)
CI (CP)	11	O	HIGHER CLASS I (CP)
ENBL T1	12	I	ENBL T1
MACLR	13	I	MACLR
TOP PRIORITY	14	O	CLASS TOP PRIORITY
CIII EN	15	O	CLASS III ENBL (IOCA)
CII EN	16	I	CP CLASS II ENBL

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CI EN	17	I	CP CLASS I ENBL
CI INTR	18	O	MICRO CLASS I INTRPT
LD CII STK	19	I	LD CLASS II STACK
LD CI STK	20	I	LD CLASS I STACK
COMM PNDNG	21	I	COMMAND PENDING
EN CIII LTC	22	O	ENBL LOAD CLASS III LATCH
CLK CII STK	23	O	CLOCK CLASS II STACK
MAE IN	24	I	MAE IN
EN CIII INTR	26	I	ENBL CLASS III INTERRUPT (CP)
T SCAN	27	I	T SCAN (TR=G)
SET CH REQ	28	O	SET CHAIN REQ + CHAIN REQ
IO CLK	29	I	IO CLK
DATA/ADRS TO	30	O	DATA/ADRS TO
PARITY ERR	32	O	PARITY ERROR
CLASS INTR	33	O	MICRO CLASS INTR
T/R	34	I	T/R (L = IOC RCV)
CIII INTR	35	O	MICRO CLASS III INTERRUPT
MPC MODE	36	I	MPC MODE
IN INTR UP	37	I	INHIBIT INTERRUPT UPDATE
LOOPBACK	38	I	LOOPBACK

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
PROTECT ERR	39	O	PROTECT ERROR
GRANTED	41	I	GRANTED
ABORT	42	I	ABORT
B/DESTN 3	43	I/O	IOC SCE B/DESTN BUS 3
B/DESTN 2	44	I/O	IOC SCE B/DESTN BUS 2
B/DESTN 1	45	I/O	IOC SCE B/DESTN BUS 1
B/DESTN 0	46	I/O	IOC SCE B/DESTN BUS 0
ADRS TO	47	I	ADR TO
DATA TO	48	I	DATA TO
CLK MDR	49	I	CLK MDR
PARITY	50	I	PARITY
RD DATA 07	52	I/O	CB RD DATA 07
RD DATA 06	53	I/O	CB RD DATA 06
RD DATA 05	54	I/O	CB RD DATA 05
RD DATA 04	55	I/O	CB RD DATA 04
RD DATA 03	56	I/O	CB RD DATA 03
RD DATA 02	57	I/O	CB RD DATA 02
RD DATA 01	58	I/O	CB RD DATA 01
RD DATA 00	59	I/O	CB RD DATA 00
CIII HIT	61	O	CLASS III HIT

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
EN CM WR	62	O	ENBL CM WRITE
T4	63	O	(ESA + DMA) X T4
EI	64	O	EI
WR DATA HIT	65	O	WRITE DATA HIT
NO PROTECT	66	O	NO PROTECT
BYTE MODE	67	O	BYTE MODE
SET ADRS 0	68	O	SET ADDRESS BIT 0
DUAL1ST PASS	69	O	DUAL FIRST PASS $\overline{\text{DSC RCW}}$
SLOW	70	O	SLOW CLK + DMA + ESA
PAGE SEL 00	71	O	PAGE SEL 00
S/D LTC	72	I	DESTN S/D LATCH
CH INTR HIT	73	I	CHAIN INTRPT HIT
LD CTRL WD	74	I	LOAD DL CONTROL WORD
DATA HIT	76	I	DATA HIT
ERROR	77	O	ERROR
CLK CI STK	78	O	CLK CLASS I STACK
CHAN 1	79	I	CHAN SEL 1 (ACK)
CHAN 0	80	I	CHAN SEL 0 (ACK)
CHAN 2	82	I	CHAN SEL 2 (ACK)
CHAN 3	83	I	CHAN SEL 3 (ACK)

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6.4.7 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MDR + T3	84	I	SOURCE MDR + T3 DATA
IR BIT 8	85	I	IOC MICRO IR BIT 8
FAULT	86	O	I/O INST FAULT
B/DESTN 7	87	I/O	IOC SCE B/DESTN BUS 7
B/DESTN 6	88	I/O	IOC SCE B/DESTN BUS 6
B/DESTN 5	89	I/O	IOC SCE B/DESTN BUS 5
B/DESTN 4	91	I/O	IOC SCE B/DESTN BUS 4
BUFF COMP	92	I	BUFFER COMPLETE
BIT 7	93	I/O	IOC BUS BIT 7
BIT 6	94	I/O	IOC BUS BIT 6
BIT 5	95	I/O	IOC BUS BIT 5
BIT 4	96	I/O	IOC BUS BIT 4
BIT 3	97	I/O	IOC BUS BIT 3
BIT 2	98	I/O	IOC BUS BIT 2
BIT 1	99	I/O	IOC BUS BIT 1
BIT 0	100	I/O	IOC BUS BIT 0

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N720)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to	B to	Min	Max		
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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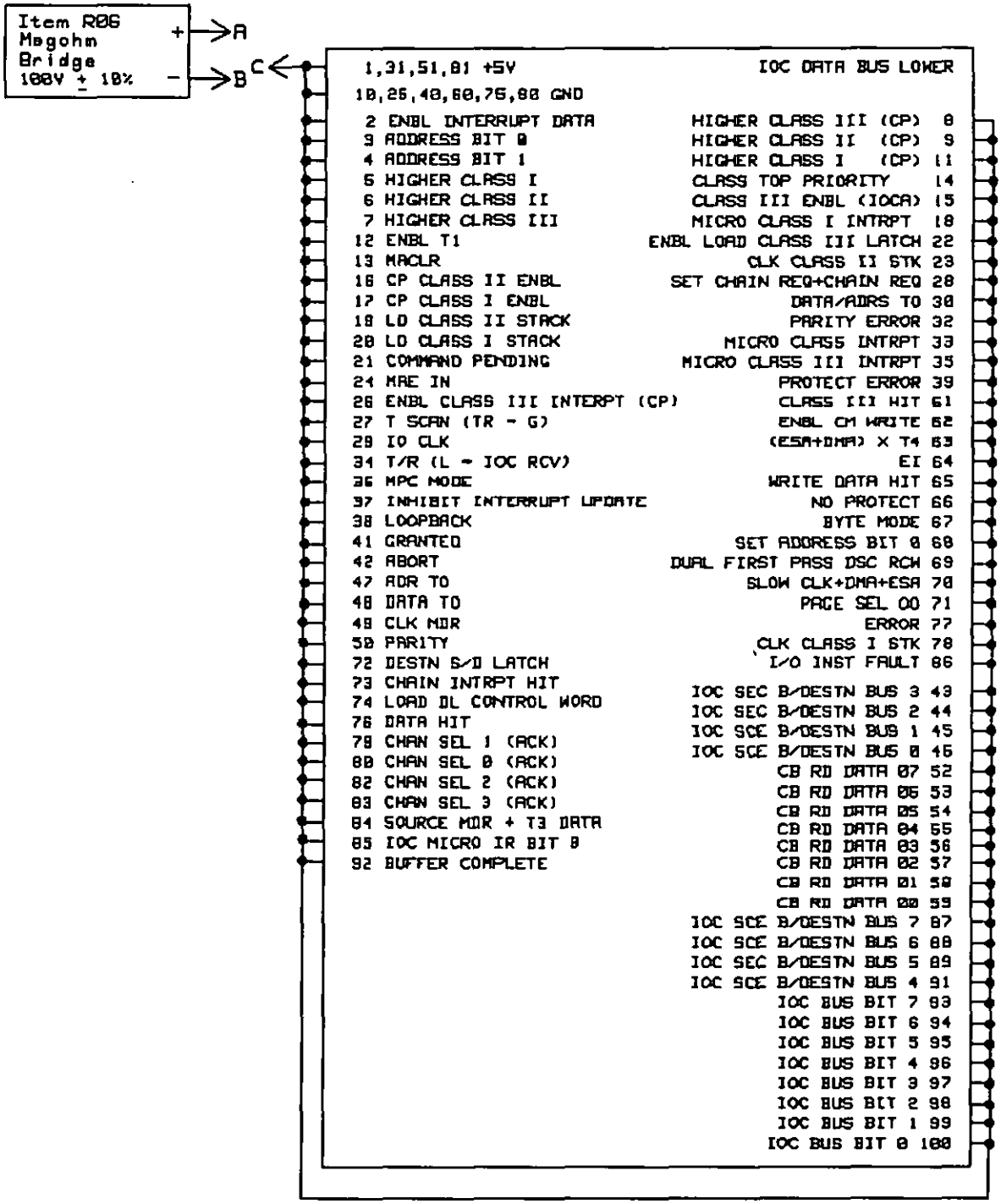


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to	B to	Min	Max		
B1	Continuity, ground	10	25	0	1	Ohm	1-5
B2	Continuity, ground	10	40				
B3	Continuity, ground	60	75				
B4	Continuity, ground	60	90				
B5	Continuity, + 5V	1	31				
B6	Continuity, + 5V	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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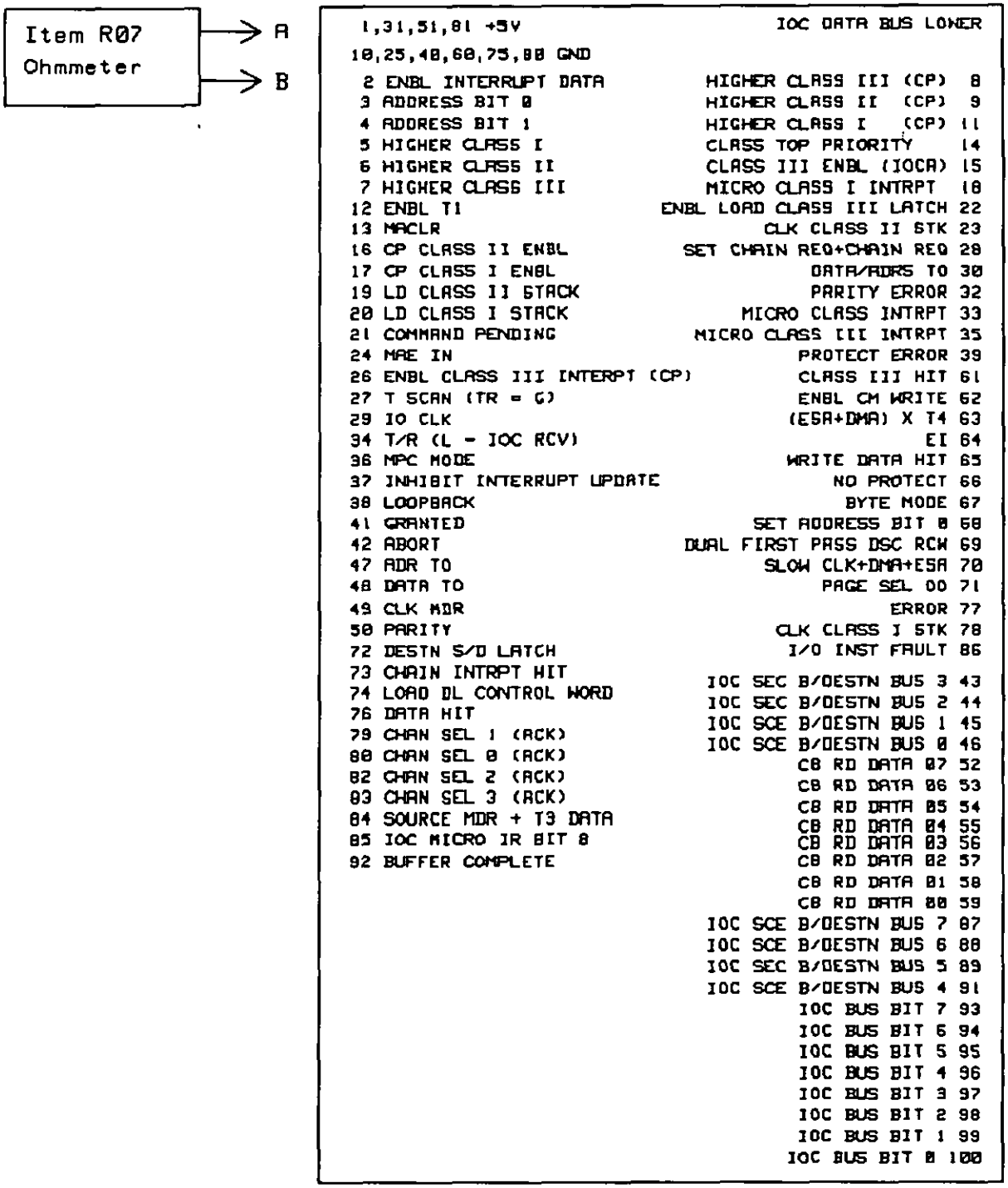


FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	1.79 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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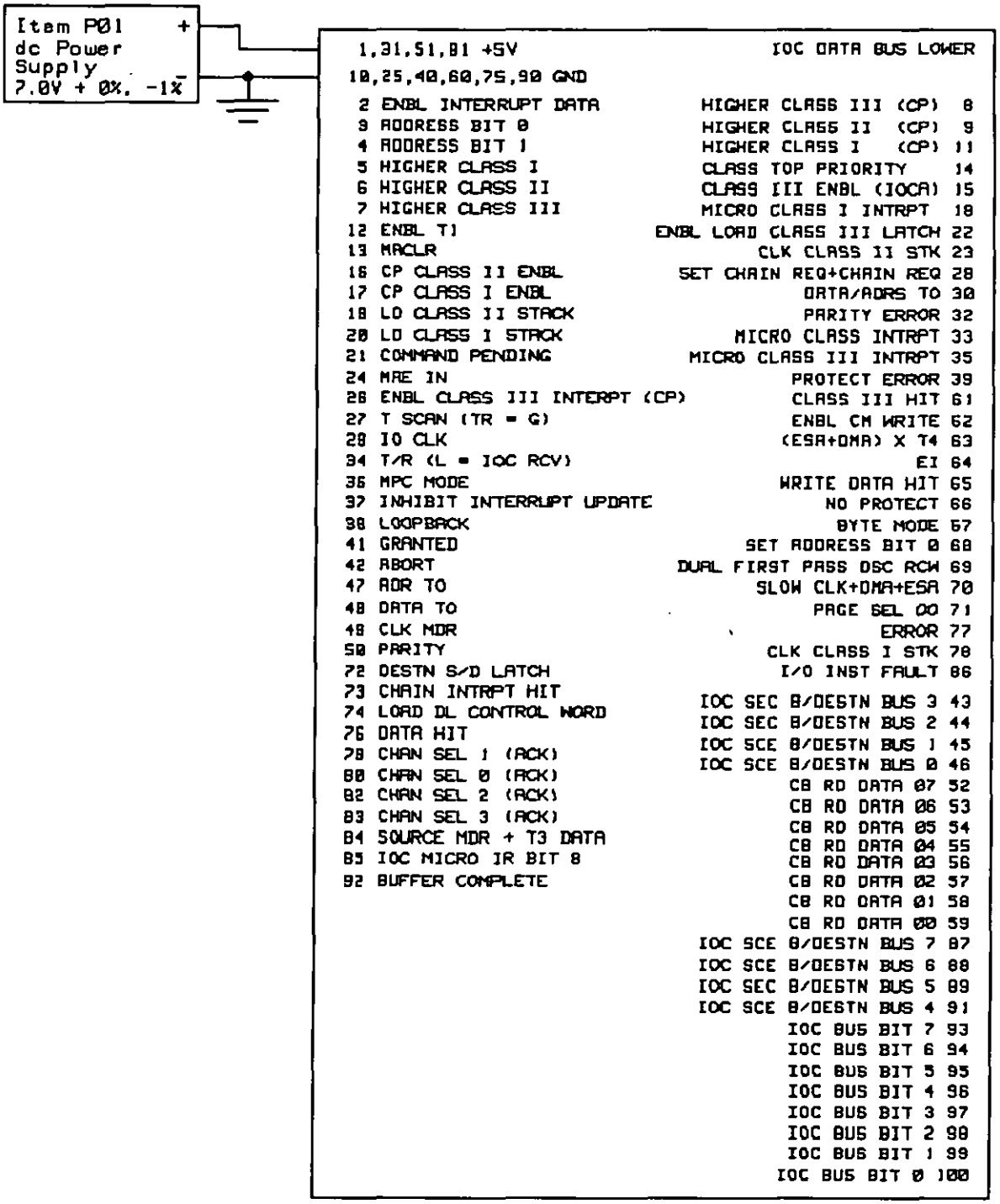


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D1	EN INTR DATA	2	5.5	0.0	1.2	mA	1-3
D2	ADRS 0	3	5.5	0.0	1.0	mA	1-3
D3	ADRS 1	4	5.5	0.0	1.0	mA	1-3
D4	CI	5	5.5	-0.12	1.11	mA	1-3
D5	CII	6	5.5	-0.12	1.11	mA	1-3
D6	CIII	7	5.5	-0.12	2.11	mA	1-3
D7	ENBL T1	12	5.5	0.0	0.1	mA	1-3
D8	MACLR	13					
D9	CII EN	16					
D10	CI EN	17					
D11	LD CII STK	19					
D12	LD CI STK	20	5.5	0.0	0.1	mA	1-3
D13	COMM PNDNG	21	5.5	0.0	1.0	mA	1-3
D14	MAE IN	24	5.5	0.0	0.1	mA	1-3
D15	EN CIII INTR	26					
D16	T SCAN	27	5.5	0.0	0.1	mA	1-3
D17	IO CLK	29	5.5	0.0	11.2	mA	1-3
D18	T/R	34	5.5	0.0	1.2	mA	1-3
D19	MPC MODE	36	5.5	0.0	2.9	mA	1-3
D20	IN INTR UP	37	5.5	0.0	0.2	mA	1-3
D21	LOOPBACK	38	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D22	GRANTED	41	5.5	0.0	0.1	mA	1-3
D23	ABORT	42					
D24	ADRS TO	47					
D25	DATA TO	48	5.5	0.0	0.1	mA	1-3
D26	CLK MDR	49	5.5	0.0	0.2	mA	1-3
D27	PARITY	50	5.5	0.0	0.1	mA	1-3
D28	S/D LTC	72	5.5	0.0	0.1	mA	1-3
D29	CH INTR HIT	73	5.5	0.0	0.3	mA	1-3
D30	LD CTRL WD	74	5.5	0.0	0.1	mA	1-3
D31	DATA HIT	76	5.5	0.0	0.3	mA	1-3
D32	CHAN 1	79	5.5	0.0	0.1	mA	1-3
D33	CHAN 0	80					
D34	CHAN 2	82					
D35	CHAN 3	83	5.5	0.0	0.1	mA	1-3
D36	MDR + T3	84	5.5	0.0	0.2	mA	1-3
D37	IR BIT 8	85	5.5	0.0	0.2	mA	1-3
D38	BUFF COMP	92	5.5	0.0	0.1	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D38
Power supply current limit	1.49 A	NA	D1-D38
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D38
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D38 D1-D38
Input current measurement	NA	± 60.0 uA	D1
	NA	± 50.0 uA	D2,D3
	NA	± 61.5 uA	D4,D5
	NA	± 111.0 uA	D6
	NA	± 5.0 uA	D7-D12
	NA	± 50.0 uA	D13
	NA	± 5.0 uA	D14-D16
	NA	± 560.0 uA	D17
	NA	± 60.0 uA	D18
	NA	± 145.0 uA	D19
	NA	± 10.0 uA	D20
	NA	± 5.0 uA	D21-D25
	NA	± 10.0 uA	D26
	NA	± 5.0 uA	D27,D28
	NA	± 15.0 uA	D29
	NA	± 5.0 uA	D30
	NA	± 15.0 uA	D31
	NA	± 5.0 uA	D32-D35
	NA	± 10.0 uA	D36,D37
	NA	± 5.0 uA	D38

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	CIII (CP)	8	0.0	-100.0	-40.0	mA	1-4
E2	CII (CP)	9	0.0	-130.0	-30.0	mA	1-4
E3	CI (CP)	11	0.0	-130.0	-30.0	mA	1-4
E4	TOP PRIORITY	14	0.0	-150.0	-60.0	mA	1-4
E5	CIII EN	15					
E6	CI INTR	18					
E7	EN CIII LTC	22	0.0	-150.0	-60.0	mA	1-4
E8	CLK CII STK	23	0.0	-100.0	-40.0	mA	1-4
E9	SET CH REQ	28	0.0	-150.0	-60.0	mA	1-4
E10	DATA/ADRS TO	30					
E11	PARITY ERR	32					
E12	CLASS INTR	33	0.0	-150.0	-60.0	mA	1-4
E13	CIII INTR	35	0.0	-260.0	-30.0	mA	1-4
E14	PROTECT ERR	39	0.0	-150.0	-60.0	mA	1-4
E15	B/DESTN 3	43					
E16	B/DESTN 2	44					
E17	B/DESTN 1	45					
E18	B/DESTN 0	46	0.0	-150.0	-60.0	mA	1-4
E19	RD DATA 07	52	0.0	-325.0	-100.0	mA	1-4
E20	RD DATA 06	53	0.0				
E21	RD DATA 05	54	0.0	-325.0	-100.0	mA	1-4

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Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	RD DATA 04	55	0.0	-325.0	-100.0	mA	1-4
E23	RD DATA 03	56					
E24	RD DATA 02	57					
E25	RD DATA 01	58					
E26	RD DATA 00	59	0.0	-325.0	-100.0	mA	1-4
E27	CIII HIT	61	0.0	-150.0	-60.0	mA	1-4
E28	EN CM WR	62					
E29	T4	63					
E30	EI	64	0.0	-150.0	-60.0	mA	1-4
E31	WR DATA HIT	65	0.0	-100.0	-20.0	mA	1-4
E32	NO PROTECT	66					
E33	BYTE MODE	67					
E34	SET ADRS 0	68	0.0	-100.0	-20.0	mA	1-4
E35	DUAL1ST PASS	69	0.0	-150.0	-60.0	mA	1-4
E36	SLOW	70					
E37	PAGE SEL 00	71					
E38	ERROR	77	0.0	-150.0	-60.0	mA	1-4
E39	CLK CI STK	78	0.0	-100.0	-40.0	mA	1-4
E40	FAULT	86	0.0	-260.0	-30.0	mA	1-4
E41	B/DESTN 7	87	0.0	-150.0	-60.0	mA	1-4
E42	B/DESTN 6	88	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E43	B/DESTN 5	89	0.0	-150.0	-60.0	mA	1-4
E44	B/DESTN 4	91					
E45	BIT 7	93					
E46	BIT 6	94					
E47	BIT 5	95					
E48	BIT 4	96					
E49	BIT 3	97	0.0	-150.0	-60.0	mA	1-4
E50	BIT 2	98	0.0	-155.0	-65.8	mA	1-4
E51	BIT 1	99	0.0	-155.0	-65.8	mA	1-4
E52	BIT 0	100	0.0	-150.0	-60.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E52
Power supply current limit	1.49 A	NA	E1-E52
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E52

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:22Z
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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	E1-E52
low, (VL)	0.0 V	± 100 mV	E1-E52
Output current measurement	NA	± 3.0 mA	E1
	NA	± 5.0 mA	E2,E3
	NA	± 4.50 mA	E4-E7
	NA	± 3.0 mA	E8
	NA	± 4.50 mA	E9-E12
	NA	± 11.5 mA	E13
	NA	± 4.50 mA	E14-E18
	NA	± 11.2 mA	E19-E26
	NA	± 4.50 mA	E27-E30
	NA	± 4.0 mA	E31-E34
	NA	± 4.50 mA	E35-E38
	NA	± 3.0 mA	E39
	NA	± 11.5 mA	E40
	NA	± 4.50 mA	E41-E49
	NA	± 4.46 mA	E50,E51
	NA	± 4.50 mA	E52

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	EN INTR DATA	2	2.4	-2.25	0.07	mA	1-3
F2	ADRS 0	3	2.4	-0.25	0.03	mA	1-3
F3	ADRS 1	4	2.4	-0.25	0.03	mA	1-3
F4	CI	5	2.4	-5.68	-2.85	mA	1-3
F5	CII	6	2.4	-5.68	-2.85	mA	1-3
F6	CIII	7	2.4	-7.68	-2.8	mA	1-3
F7	ENBL T1	12	2.4	-0.6	0.02	mA	1-3
F8	MACLR	13					
F9	CII EN	16					
F10	CI EN	17					
F11	LD CII STK	19					
F12	LD CI STK	20	2.4	-0.6	0.02	mA	1-3
F13	COMM PNDNG	21	2.4	-2.0	0.05	mA	1-3
F14	MAE IN	24	2.4	-0.6	0.02	mA	1-3
F15	EN CIII INTR	26					
F16	T SCAN	27	2.4	-0.6	0.02	mA	1-3
F17	IO CLK	29	2.4	-21.4	0.57	mA	1-3
F18	T/R	34	2.4	-2.8	0.09	mA	1-3
F19	MPC MODE	36	2.4	-8.45	0.26	mA	1-3
F20	IN INTR UP	37	2.4	-0.8	0.04	mA	1-3
F21	LOOPBACK	38	2.4	-0.6	0.02	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	GRANTED	41	2.4	-0.6	0.02	mA	1-3
F23	ABORT	42	2.4	-0.6	0.02	mA	1-3
F24	B/DESTN 3	43	2.4	-0.45	0.09	mA	1-3
F25	B/DESTN 2	44					
F26	B/DESTN 1	45					
F27	B/DESTN 0	46	2.4	-0.45	0.09	mA	1-3
F28	ADRS TO	47	2.4	-1.8	0.02	mA	1-3
F29	DATA TO	48	2.4	-1.8	0.02	mA	1-3
F30	CLK MDR	49	2.4	-0.8	0.04	mA	1-3
F31	PARITY	50	2.4	-0.6	0.02	mA	1-3
F32	RD DATA 07	52	2.4	-0.25	0.07	mA	1-3
F33	RD DATA 06	53					
F34	RD DATA 05	54					
F35	RD DATA 04	55					
F36	RD DATA 03	56					
F37	RD DATA 02	57					
F38	RD DATA 01	58					
F39	RD DATA 00	59	2.4	-0.25	0.07	mA	1-3
F40	S/D LTC	72	2.4	-0.2	0.02	mA	1-3
F41	CH INTR HIT	73	2.4	-1.4	0.06	mA	1-3
F42	LD CTRL WD	74	2.4	-0.2	0.02	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F43	DATA HIT	76	2.4	-1.2	0.06	mA	1-3
F44	CHAN 1	79	2.4	-0.6	0.02	mA	1-3
F45	CHAN 0	80					
F46	CHAN 2	82					
F47	CHAN 3	83	2.4	-0.6	0.02	mA	1-3
F48	MDR + T3	84	2.4	-1.2	0.04	mA	1-3
F49	IR BIT 8	85	2.4	-1.2	0.04	mA	1-3
F50	B/DESTN 7	87	2.4	-0.45	0.09	mA	1-3
F51	B/DESTN 6	88					
F52	B/DESTN 5	89					
F53	B/DESTN 4	91	2.4	-0.45	0.09	mA	1-3
F54	BUFF COMP	92	2.4	-0.2	0.02	mA	1-3
F55	BIT 7	93	2.4	-0.25	0.07	mA	1-3
F56	BIT 6	94	2.4	-0.25	0.07	mA	1-3
F57	BIT 5	95	2.4	-0.85	0.09	mA	1-3
F58	BIT 4	96	2.4	-0.85	0.09	mA	1-3
F59	BIT 3	97	2.4	-0.25	0.07	mA	1-3
F60	BIT 2	98	2.4	-5.68	-2.8	mA	1-3
F61	BIT 1	99	2.4	-5.68	-2.8	mA	1-3
F62	BIT 0	100	2.4	-0.25	0.07	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:22Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F62
Power supply current limit	1.49 A	NA	F1-F62
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F62
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F62 F1-F62
Input current measurement	NA	± 116.0 uA	F1
	NA	± 14.0 uA	F2,F3
	NA	± 141.0 uA	F4,F5
	NA	± 244.0 uA	F6
	NA	± 31.0 uA	F7-F12
	NA	± 102.0 uA	F13
	NA	± 31.0 uA	F14-F16
	NA	± 1100.0 uA	F17
	NA	± 144.0 uA	F18
	NA	± 440.0 uA	F19
	NA	± 42.0 uA	F20
	NA	± 31.0 uA	F21-F23
	NA	± 27.0 uA	F24-F27
	NA	± 91.0 uA	F28,F29
	NA	± 42.0 uA	F30
	NA	± 31.0 uA	F31
	NA	± 16.0 uA	F32-F39
	NA	± 11.0 uA	F40
	NA	± 73.0 uA	F41
	NA	± 11.0 uA	F42
	NA	± 63.0 uA	F43

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 31.0 uA	F44-F47
	NA	± 62.0 uA	F48,F49
	NA	± 27.0 uA	F50-F53
	NA	± 11.0 uA	F54
	NA	± 16.0 uA	F55,F56
	NA	± 47.0 uA	F57,F58
	NA	± 16.0 uA	F59
	NA	± 144.0 uA	F60,F61
	NA	± 16.0 uA	F62

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	EN INTR DATA	2	0.4	-2.25	0.07	mA	1-3
G2	ADRS 0	3	0.4	-0.25	0.03	mA	1-3
G3	ADRS 1	4	0.4	-0.25	0.03	mA	1-3
G4	CI	5	0.4	-5.68	-2.85	mA	1-3
G5	CII	6	0.4	-5.68	-2.85	mA	1-3
G6	CIII	7	0.4	-7.68	-2.8	mA	1-3
G7	ENBL T1	12	0.4	-0.6	0.02	mA	1-3
G8	MACLR	13					
G9	CII EN	16					
G10	CI EN	17					
G11	LD CII STK	19					
G12	LD CI STK	20	0.4	-0.6	0.02	mA	1-3
G13	COMM PNDNG	21	0.4	-2.0	0.05	mA	1-3
G14	MAE IN	24	0.4	-0.6	0.02	mA	1-3
G15	EN CIII INTR	26					
G16	T SCAN	27	0.4	-0.6	0.02	mA	1-3
G17	IO CLK	29	0.4	-21.4	0.57	mA	1-3
G18	T/R	34	0.4	-2.8	0.09	mA	1-3
G19	MPC MODE	36	0.4	-8.45	0.26	mA	1-3
G20	IN INTR UP	37	0.4	-0.8	0.04	mA	1-3
G21	LOOPBACK	38	0.4	-0.6	0.02	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	GRANTED	41	0.4	-0.6	0.02	mA	1-3
G23	ABORT	42	0.4	-0.6	0.02	mA	1-3
G24	B/DESTN 3	43	0.4	-0.45	0.09	mA	1-3
G25	B/DESTN 2	44					
G26	B/DESTN 1	45					
G27	B/DESTN 0	46	0.4	-0.45	0.09	mA	1-3
G28	ADRS TO	47	0.4	-1.8	0.02	mA	1-3
G29	DATA TO	48	0.4	-1.8	0.02	mA	1-3
G30	CLK MDR	49	0.4	-0.8	0.04	mA	1-3
G31	PARITY	50	0.4	-0.6	0.02	mA	1-3
G32	RD DATA 07	52	0.4	-0.25	0.07	mA	1-3
G33	RD DATA 06	53					
G34	RD DATA 05	54					
G35	RD DATA 04	55					
G36	RD DATA 03	56					
G37	RD DATA 02	57					
G38	RD DATA 01	58					
G39	RD DATA 00	59	0.4	-0.25	0.07	mA	1-3
G40	S/D LTC	72	0.4	-0.2	0.02	mA	1-3
G41	CH INTR HIT	73	0.4	-1.4	0.06	mA	1-3
G42	LD CTRL WD	74	0.4	-0.2	0.02	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:22Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G43	DATA HIT	76	0.4	-1.2	0.06	mA	1-3
G44	CHAN 1	79	0.4	-0.6	0.02	mA	1-3
G45	CHAN 0	80					
G46	CHAN 2	82					
G47	CHAN 3	83	0.4	-0.6	0.02	mA	1-3
G48	MDR + T3	84	0.4	-1.2	0.04	mA	1-3
G49	IR BIT 8	85	0.4	-1.2	0.04	mA	1-3
G50	B/DESTN 7	87	0.4	-0.45	0.09	mA	1-3
G51	B/DESTN 6	88					
G52	B/DESTN 5	89					
G53	B/DESTN 4	91	0.4	-0.45	0.09	mA	1-3
G54	BUFF COMP	92	0.4	-0.2	0.02	mA	1-3
G55	BIT 7	93	0.4	-0.25	0.07	mA	1-3
G56	BIT 6	94	0.4	-0.25	0.07	mA	1-3
G57	BIT 5	95	0.4	-0.85	0.09	mA	1-3
G58	BIT 4	96	0.4	-0.85	0.09	mA	1-3
G59	BIT 3	97	0.4	-0.25	0.07	mA	1-3
G60	BIT 2	98	0.4	-5.68	-2.8	mA	1-3
G61	BIT 1	99	0.4	-5.68	-2.8	mA	1-3
G62	BIT 0	100	0.4	-0.25	0.07	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 20260-20T18:22Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G62
Power supply current limit	1.49 A	NA	G1-G62
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G62
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G62 G1-G62
Input current measurement	NA	± 116.0 uA	G1
	NA	± 14.0 uA	G2,G3
	NA	± 141.0 uA	G4,G5
	NA	± 244.0 uA	G6
	NA	± 31.0 uA	G7-G12
	NA	± 102.0 uA	G13
	NA	± 31.0 uA	G14-G16
	NA	± 1100.0 uA	G17
	NA	± 144.0 uA	G18
	NA	± 440.0 uA	G19
	NA	± 42.0 uA	G20
	NA	± 31.0 uA	G21-G23
	NA	± 27.0 uA	G24-G27
	NA	± 91.0 uA	G28,G29
	NA	± 42.0 uA	G30
	NA	± 31.0 uA	G31
	NA	± 16.0 uA	G32-G39
	NA	± 11.0 uA	G40
	NA	± 73.0 uA	G41
	NA	± 11.0 uA	G42

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 63.0 uA	G43
	NA	± 31.0 uA	G44-G47
	NA	± 62.0 uA	G48,G49
	NA	± 27.0 uA	G50-G53
	NA	± 11.0 uA	G54
	NA	± 16.0 uA	G55,G56
	NA	± 47.0 uA	G57,G58
	NA	± 16.0 uA	G59
	NA	± 144.0 uA	G60,G61
	NA	± 16.0 uA	G62

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 202610-20T18:22Z
Check the source to verify that this is the current version before use.

TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	CIII (CP)	8	-0.4	2.4	4.5	V	1-4
H2	CII (CP)	9	-0.16	2.4	4.5	V	1-4
H3	CI (CP)	11	-0.16	2.4	4.5	V	1-4
H4	TOP PRIORITY	14	-0.4	2.4	4.5	V	1-4
H5	CIII EN	15					
H6	CI INTR	18					
H7	EN CIII LTC	22					
H8	CLK CII STK	23					
H9	SET CH REQ	28					
H10	DATA/ADRS TO	30					
H11	PARITY ERR	32					
H12	CLASS INTR	33	-0.4	2.4	4.5	V	1-4
H13	CIII INTR	35	-0.16	2.4	4.5	V	1-4
H14	PROTECT ERR	39	-0.4	2.4	4.5	V	1-4
H15	B/DESTN 3	43					
H16	B/DESTN 2	44					
H17	B/DESTN 1	45					
H18	B/DESTN 0	46					
H19	RD DATA 07	52					
H20	RD DATA 06	53					
H21	RD DATA 05	54	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	RD DATA 04	55	-0.4	2.4	4.5	V	1-4
H23	RD DATA 03	56					
H24	RD DATA 02	57					
H25	RD DATA 01	58					
H26	RD DATA 00	59					
H27	CIII HIT	61					
H28	EN CM WR	62					
H29	T4	63					
H30	EI	64	-0.4	2.4	4.5	V	1-4
H31	WR DATA HIT	65	-0.16	2.4	4.5	V	1-4
H32	NO PROTECT	66					
H33	BYTE MODE	67					
H34	SET ADRS 0	68	-0.16	2.4	4.5	V	1-4
H35	DUAL1ST PASS	69	-0.4	2.4	4.5	V	1-4
H36	SLOW	70					
H37	PAGE SEL 00	71					
H38	ERROR	77					
H39	CLK CI STK	78	-0.4	2.4	4.5	V	1-4
H40	FAULT	86	-0.16	2.4	4.5	V	1-4
H41	B/DESTN 7	87	-0.4	2.4	4.5	V	1-4
H42	B/DESTN 6	88	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H43	B/DESTN 5	89	-0.4	2.4	4.5	V	1-4
H44	B/DESTN 4	91					
H45	BIT 7	93					
H46	BIT 6	94					
H47	BIT 5	95					
H48	BIT 4	96					
H49	BIT 3	97					
H50	BIT 2	98					
H51	BIT 1	99					
H52	BIT 0	100	-0.4	2.4	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H52
Power supply current limit	1.49 A	NA	H1-H52
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA -0.16 mA	± 40 uA ± 16 uA ± 40 uA ± 16 uA	H1 H2,H3 H4-H12 H13

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA -0.16 mA -0.4 mA	± 40 uA ± 16 uA ± 40 uA ± 16 uA ± 40 uA	H14-H30 H31-H34 H35-H39 H40 H41-H52
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H52 H1-H52
Output voltage measurement	NA	± 105 mV	H1-H52

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

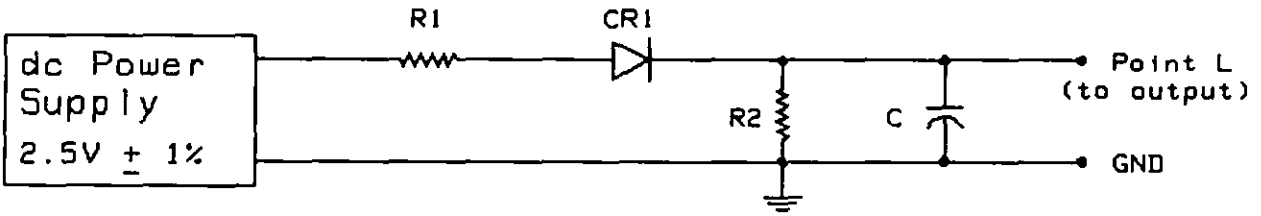
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
8,14,15,18,22,23, 28,30,32,33,39, 43,44,45,46,52, 53,54,55,56,57, 58,59,61,62,63, 64,69,70,71,77, 78,87,88,89,91, 93,94,95,96,97, 98,99,100	6200	6.0	0.5
9,11,35,65,66,67, 68,86	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	CIII (CP)	8	6.0	0.0	0.5	V	1-4
J2	CII (CP)	9	1.6	0.0	0.5	V	1-4
J3	CI (CP)	11	1.6	0.0	0.5	V	1-4
J4	TOP PRIORITY	14	6.0	0.0	0.5	V	1-4
J5	CIII EN	15					
J6	CI INTR	18					
J7	EN CIII LTC	22					
J8	CLK CII STK	23					
J9	SET CH REQ	28					
J10	DATA/ADRS TO	30					
J11	PARITY ERR	32					
J12	CLASS INTR	33	6.0	0.0	0.5	V	1-4
J13	CIII INTR	35	1.6	0.0	0.5	V	1-4
J14	PROTECT ERR	39	6.0	0.0	0.5	V	1-4
J15	B/DESTN 3	43					
J16	B/DESTN 2	44					
J17	B/DESTN 1	45					
J18	B/DESTN 0	46					
J19	RD DATA 07	52					
J20	RD DATA 06	53					
J21	RD DATA 05	54	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J22	RD DATA 04	55	6.0	0.0	0.5	V	1-4
J23	RD DATA 03	56					
J24	RD DATA 02	57					
J25	RD DATA 01	58					
J26	RD DATA 00	59					
J27	CIII HIT	61					
J28	EN CM WR	62					
J29	T4	63					
J30	EI	64	6.0	0.0	0.5	V	1-4
J31	WR DATA HIT	65	1.6	0.0	0.5	V	1-4
J32	NO PROTECT	66					
J33	BYTE MODE	67					
J34	SET ADRS 0	68	1.6	0.0	0.5	V	1-4
J35	DUAL1ST PASS	69	6.0	0.0	0.5	V	1-4
J36	SLOW	70					
J37	PAGE SEL 00	71					
J38	ERROR	77					
J39	CLK CI STK	78	6.0	0.0	0.5	V	1-4
J40	FAULT	86	1.6	0.0	0.5	V	1-4
J41	B/DESTN 7	87	6.0	0.0	0.5	V	1-4
J42	B/DESTN 6	88	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J43	B/DESTN 5	89	6.0	0.0	0.5	V	1-4
J44	B/DESTN 4	91					
J45	BIT 7	93					
J46	BIT 6	94					
J47	BIT 5	95					
J48	BIT 4	96					
J49	BIT 3	97					
J50	BIT 2	98					
J51	BIT 1	99					
J52	BIT 0	100	6.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J52
Power supply current limit	1.49 A	NA	J1-J52
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA	J1 J2,J3 J4-J12 J13

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA 6.0 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA ± 600 uA	J14-J30 J31-J34 J35-J39 J40 J41-J52
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J52 J1-J52
Output voltage measurement	NA	± 25 mV	J1-J52

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	EN INTR DATA	2	-18.0	-1.2	0.0	V	1-3
L2	ADRS 0	3	-18.0	-1.5	0.0	V	1-3
L3	ADRS 1	4	-18.0	-1.5	0.0	V	1-3
L4	ENBL T1	12	-18.0	-1.2	0.0	V	1-3
L5	MACLR	13					
L6	CII EN	16					
L7	CI EN	17					
L8	LD CII STK	19					
L9	LD CI STK	20					
L10	COMM PNDNG	21					
L11	MAE IN	24					
L12	EN CIII INTR	26					
L13	T SCAN	27					
L14	IO CLK	29					
L15	T/R	34					
L16	MPC MODE	36					
L17	IN INTR UP	37					
L18	LOOPBACK	38					
L19	GRANTED	41					
L20	ABORT	42					
L21	ADRS TO	47	-18.0	-1.2	0.0	V	1-3

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TABLE XI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L22	DATA TO	48	-18.0	-1.2	0.0	V	1-3
L23	CLK MDR	49					
L24	PARITY	50	-18.0	-1.2	0.0	V	1-3
L25	S/D LTC	72	-18.0	-1.5	0.0	V	1-3
L26	CH INTR HIT	73	-18.0	-1.2	0.0	V	1-3
L27	LD CTRL WD	74	-18.0	-1.5	0.0	V	1-3
L28	DATA HIT	76	-18.0	-1.2	0.0	V	1-3
L29	CHAN 1	79					
L30	CHAN 0	80					
L31	CHAN 2	82					
L32	CHAN 3	83					
L33	MDR + T3	84					
L34	IR BIT 8	85	-18.0	-1.2	0.0	V	1-3
L35	BUFF COMP	92	-18.0	-1.5	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L35

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Power supply current limit	NA	NA	L1-L35
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L35
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L35 L1-L35
Input voltage measurement	NA NA NA NA NA NA NA NA	± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV	L1 L2,L3 L4-L24 L25 L26 L27 L28-L34 L35

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	4.2	7.8	uF	1-4
M2	PSFC	51	4.2	7.8	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA	± 180 nF	M1,M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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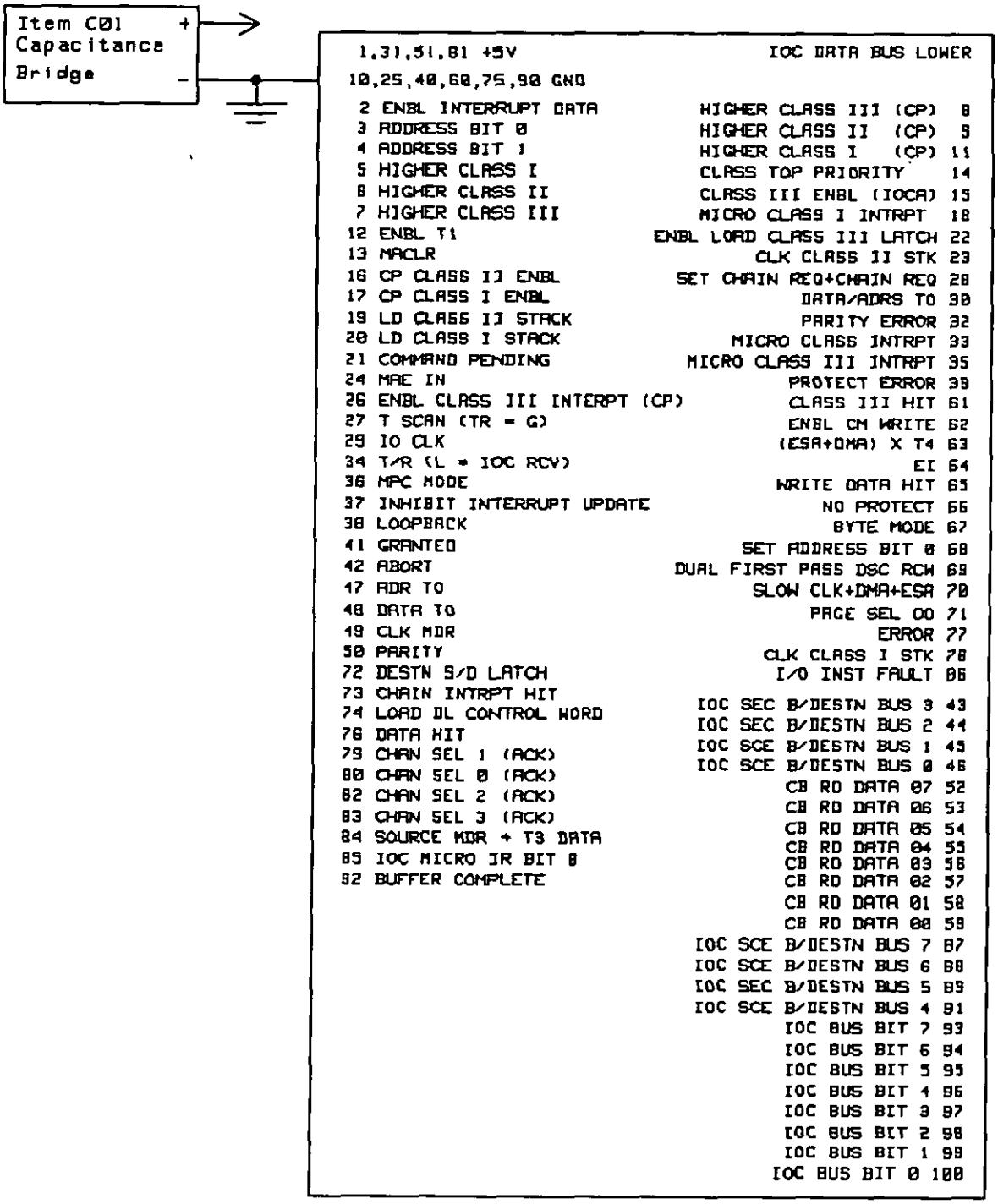


FIGURE 10. Power supply filter capacitance test circuit for table XII.

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TABLE XIII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Power supply current	1,31, 51,81	415	996	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.49 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 29 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.49 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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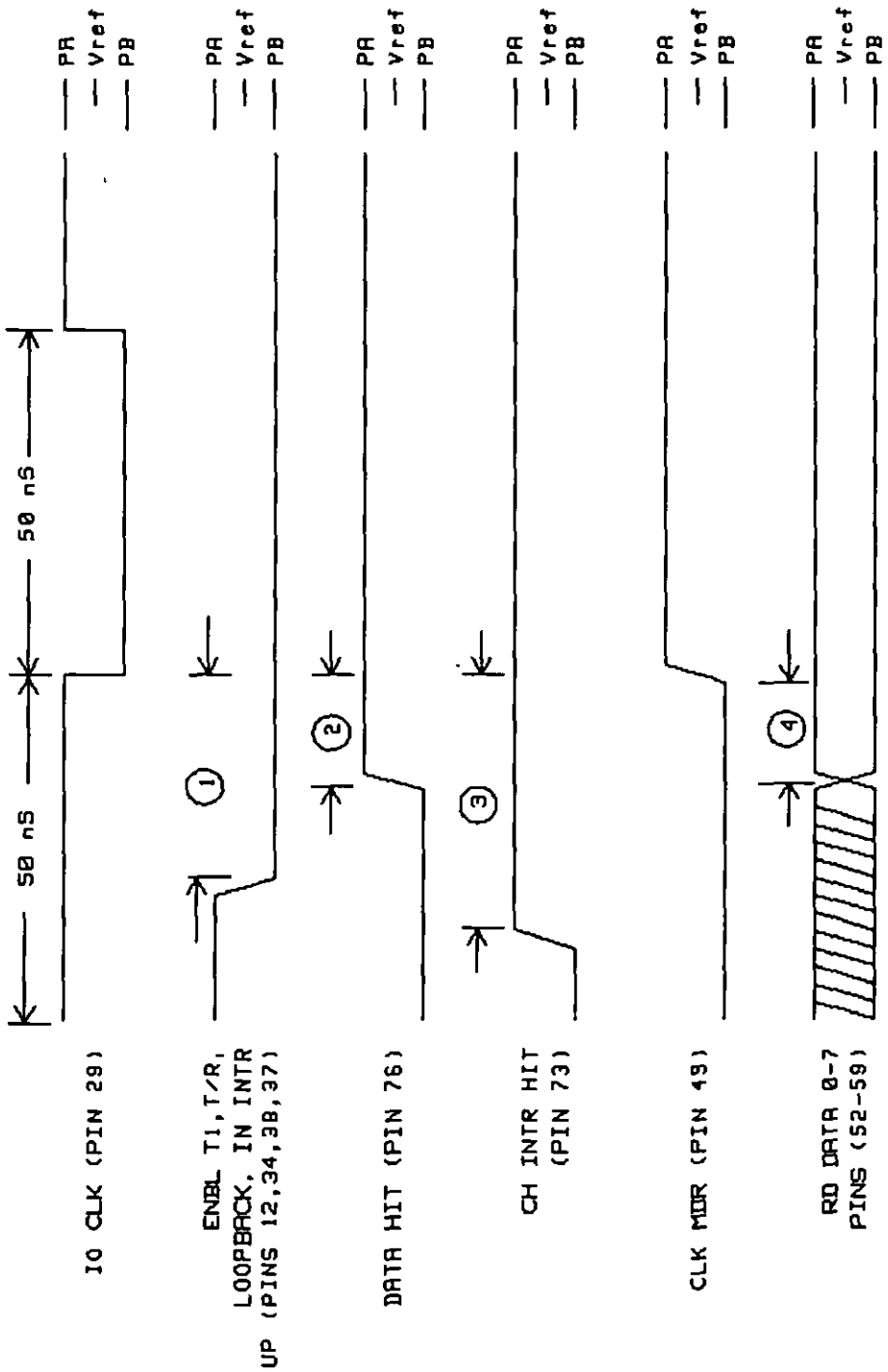


FIGURE 11. Input timing parameters for table XIV.

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NOTES:

1. (1) ENBL T1, T/R, LOOPBACK, IN INTR UP to IO CLK. Setup time, minimum 26 nS.

2. (2) DATA HIT to IO CLK. Setup time, minimum 12 nS.

3. (3) CH INTR HIT to IO CLK. Setup time, minimum 34 nS.

4. (4) RD DATA 07, RD DATA 06, RD DATA 05, RD DATA 04, RD DATA 03, RD DATA 02, RD DATA 01, RD DATA 00 to CLK MDR. Setup time, minimum 10 nS.

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TABLE XV. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.49 A	NA	Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage			
high, (VH)	2.4 V	± 100 mV	Q1-Q4
low, (VL)	0.4 V	± 100 mV	Q1-Q4
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

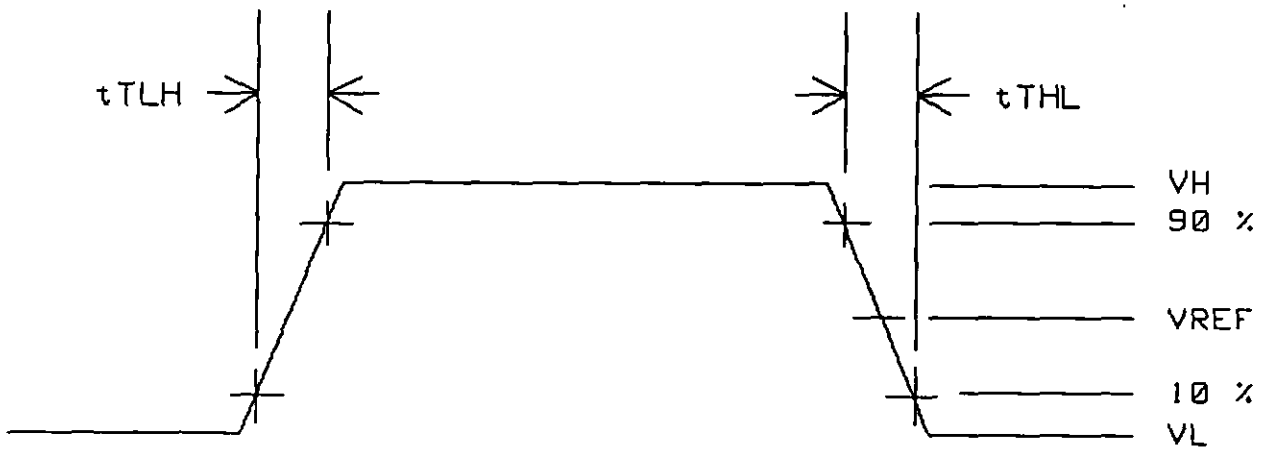
(c) The dynamic parameters are found in table XVI.

3. The load circuit for this test is specified on figure 9.

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NOTES: - Continued.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XV.

6. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MPC MODE to PRO-TECT ERR	-	-	-	23	23	23	nS	12
2	tpd	MPC MODE to PAR-ITY ERR	-	-	-	33	33	33	nS	12
2	tpd	MPC MODE to DATA/ ADRS TO	-	-	-	33	33	33	nS	12
3	tpd	MPC MODE to ERROR	-	-	-	59	59	59	nS	12
4	tpd	IO CLK to EN CIII LTC	-	-	-	55	55	55	nS	12
5	tpd	IO CLK to CIII INTR	-	-	-	36	36	36	nS	12
6	tpd	IO CLK to CIII EN	-	-	-	29	29	29	nS	12
7	tpd	IO CLK to CLK CII STK	-	-	-	17	17	17	nS	12
8	tpd	IO CLK to CLK CI STK	-	-	-	17	17	17	nS	12
9	tpd	IO CLK to CI INTR	-	-	-	45	45	45	nS	12
9	tpd	IO CLK to CLASS INTR	-	-	-	45	45	45	nS	12
10	tpd	IO CLK to CIII HIT	-	-	-	39	39	39	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	IO CLK to SET CH REQ	-	-	-	30	30	30	nS	12
12	tpd	CI to CI (CP)	-	-	-	52	52	52	nS	12
12	tpd	CII to CI (CP)	-	-	-	52	52	52	nS	12
12	tpd	ADRS 0 to CI (CP)	-	-	-	52	52	52	nS	12
12	tpd	ADRS 1 to CI (CP)	-	-	-	52	52	52	nS	12
13	tpd	CI to TOP PRIORITY	-	-	-	59	59	59	nS	12
13	tpd	CII to TOP PRI-ORITY	-	-	-	59	59	59	nS	12
13	tpd	CIII to TOP PRI-ORITY	-	-	-	59	59	59	nS	12
13	tpd	ADRS 0 to TOP PRI-ORITY	-	-	-	59	59	59	nS	12
13	tpd	ADRS 1 to TOP PRI-ORITY	-	-	-	59	59	59	nS	12
14	tpd	CIII to CIII (CP)	-	-	-	17	17	17	nS	12
15	tpd	IO CLK to DUAL1ST PASS	-	-	-	32	32	32	nS	12
15	tpd	IO CLK to EI	-	-	-	32	32	32	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
16	tpd	IO CLK to SET ADRS 0	-	-	-	58	58	58	nS	12
17	tpd	IO CLK to SLOW	-	-	-	63	63	63	nS	12
18	tpd	IO CLK to T4	-	-	-	44	44	44	nS	12
19	tpd	IO CLK to WR DATA HIT	-	-	-	58	58	58	nS	12
19	tpd	IO CLK to NO PRO-TECT	-	-	-	58	58	58	nS	12
19	tpd	IO CLK to BYTE MODE	-	-	-	58	58	58	nS	12
20	tpd	IO CLK to PAGE SEL 00	-	-	-	70	70	70	nS	12
21	tpd	IO CLK to EN CM WR	-	-	-	51	51	51	nS	12
22	tpd	IO CLK to B/DESTN 3	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 2	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 1	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 0	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 7	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 6	-	-	-	147	147	147	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
22	tpd	IO CLK to B/DESTN 5	-	-	-	147	147	147	nS	12
22	tpd	IO CLK to B/DESTN 4	-	-	-	147	147	147	nS	12
23	tpd	IO CLK to FAULT	-	-	-	36	36	36	nS	12
24	tpd	IO CLK to BIT 7	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 6	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 5	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 4	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 3	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 2	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 1	-	-	-	32	32	32	nS	12
24	tpd	IO CLK to BIT 0	-	-	-	32	32	32	nS	12
25	tsu	ENBL T1 to IO CLK	26	26	26	-	-	-	nS	12
25	tsu	T/R to IO CLK	26	26	26	-	-	-	nS	12
25	tsu	LOOPBACK to IO CLK	26	26	26	-	-	-	nS	12
25	tsu	IN INTR UP to IO CLK	26	26	26	-	-	-	nS	12
26	tsu	DATA HIT to IO CLK	12	12	12	-	-	-	nS	12

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TABLE XVI. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
27	tsu	CH INTR HIT to IO CLK	34	34	34	-	-	-	nS	12
28	tsu	RD DATA 07 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 06 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 05 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 04 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 03 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 02 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 01 to CLK MDR	10	10	10	-	-	-	nS	12
28	tsu	RD DATA 00 to CLK MDR	10	10	10	-	-	-	nS	12

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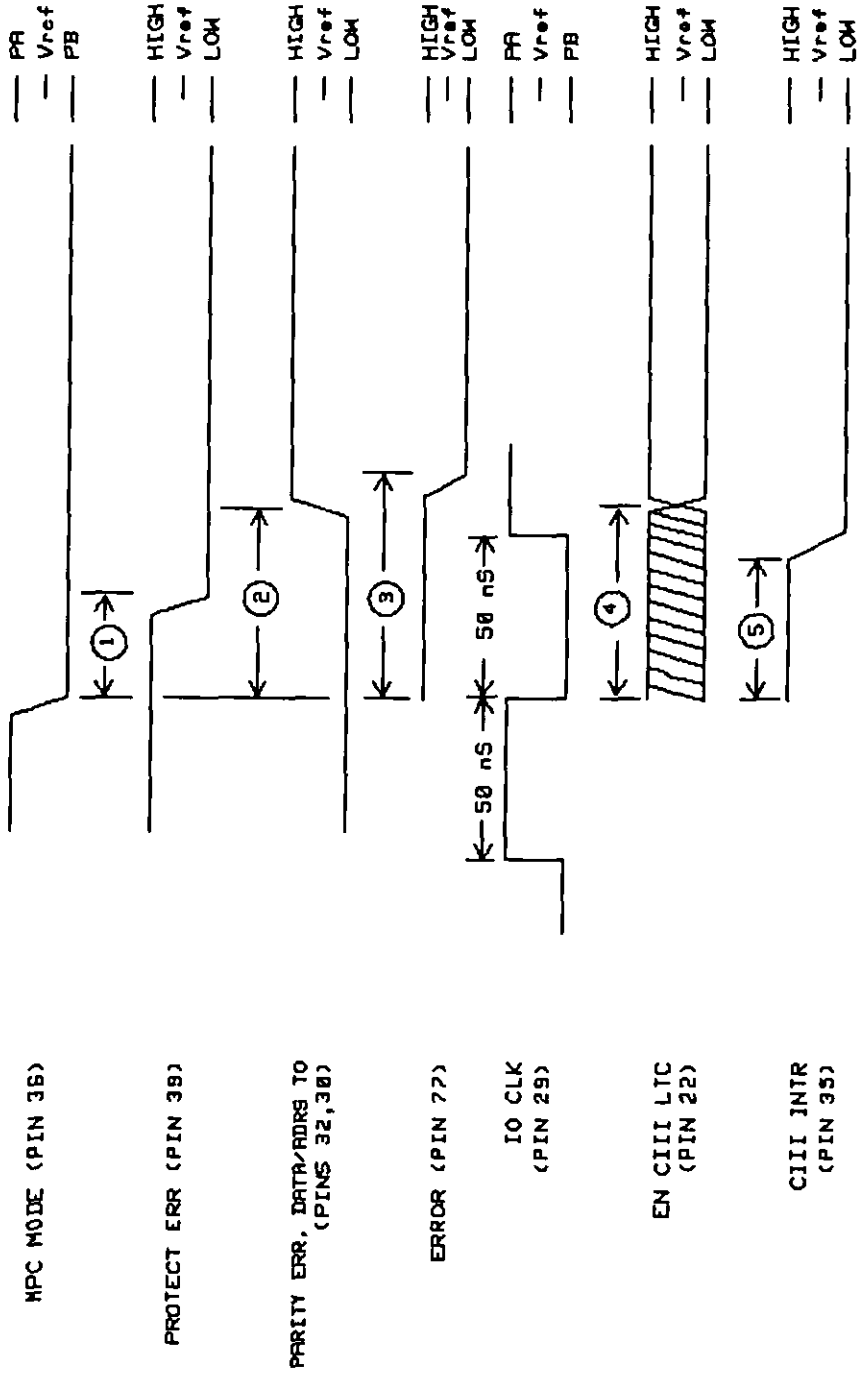


FIGURE 12a. Dynamic waveforms for table XVI.

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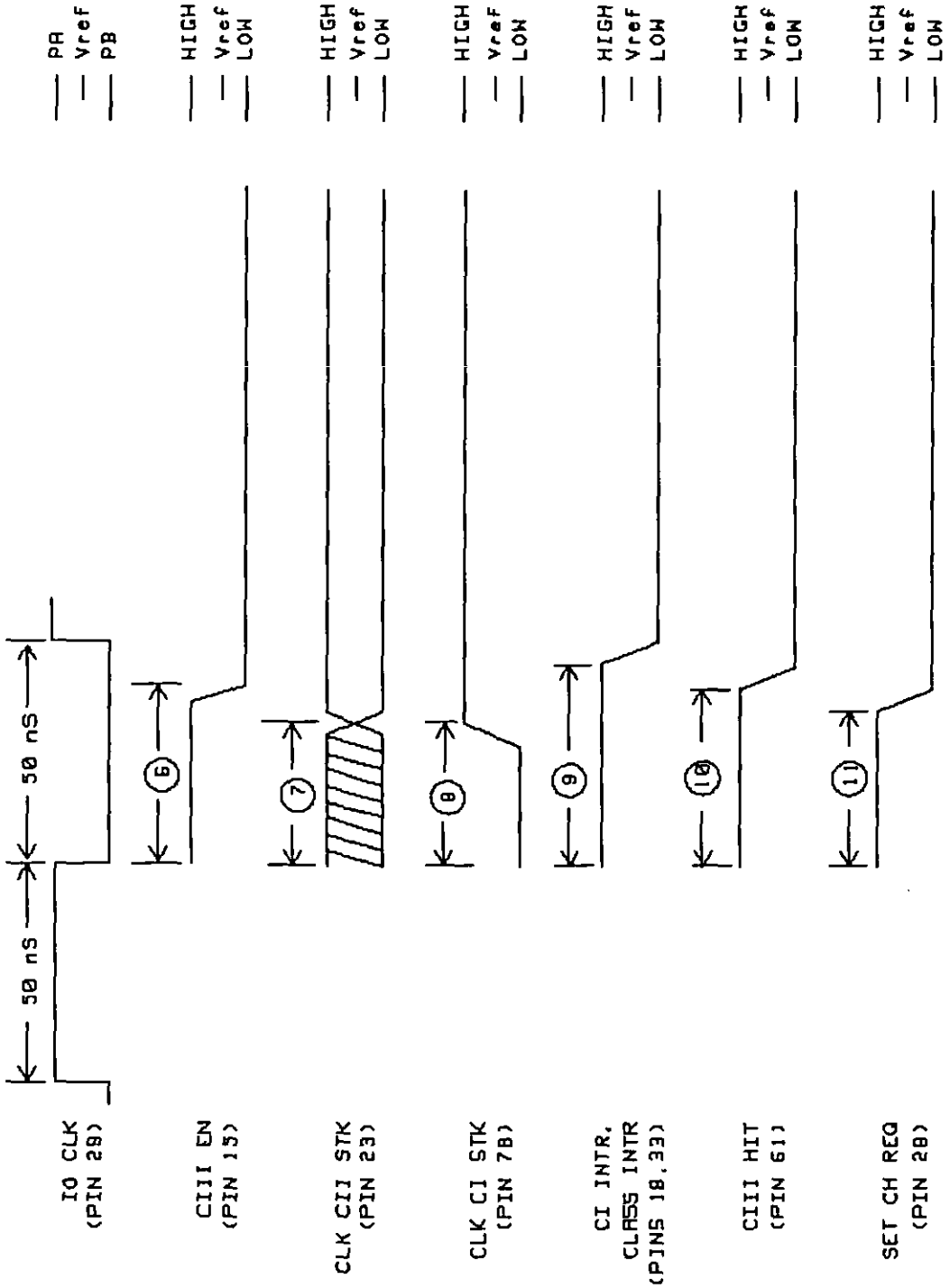


FIGURE 12b. Dynamic waveforms for table XVI.

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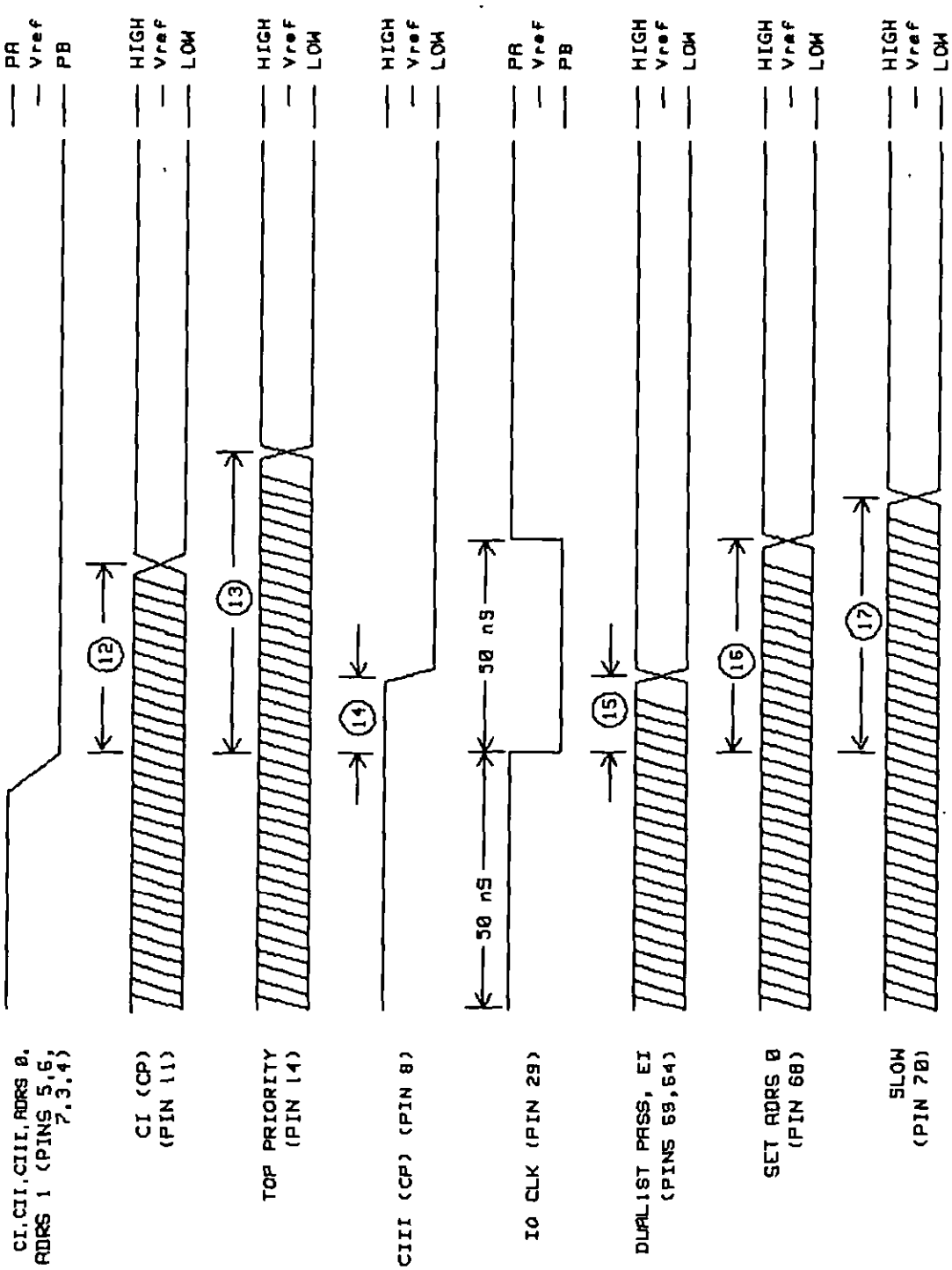


FIGURE 12c. Dynamic waveforms for table XVI.

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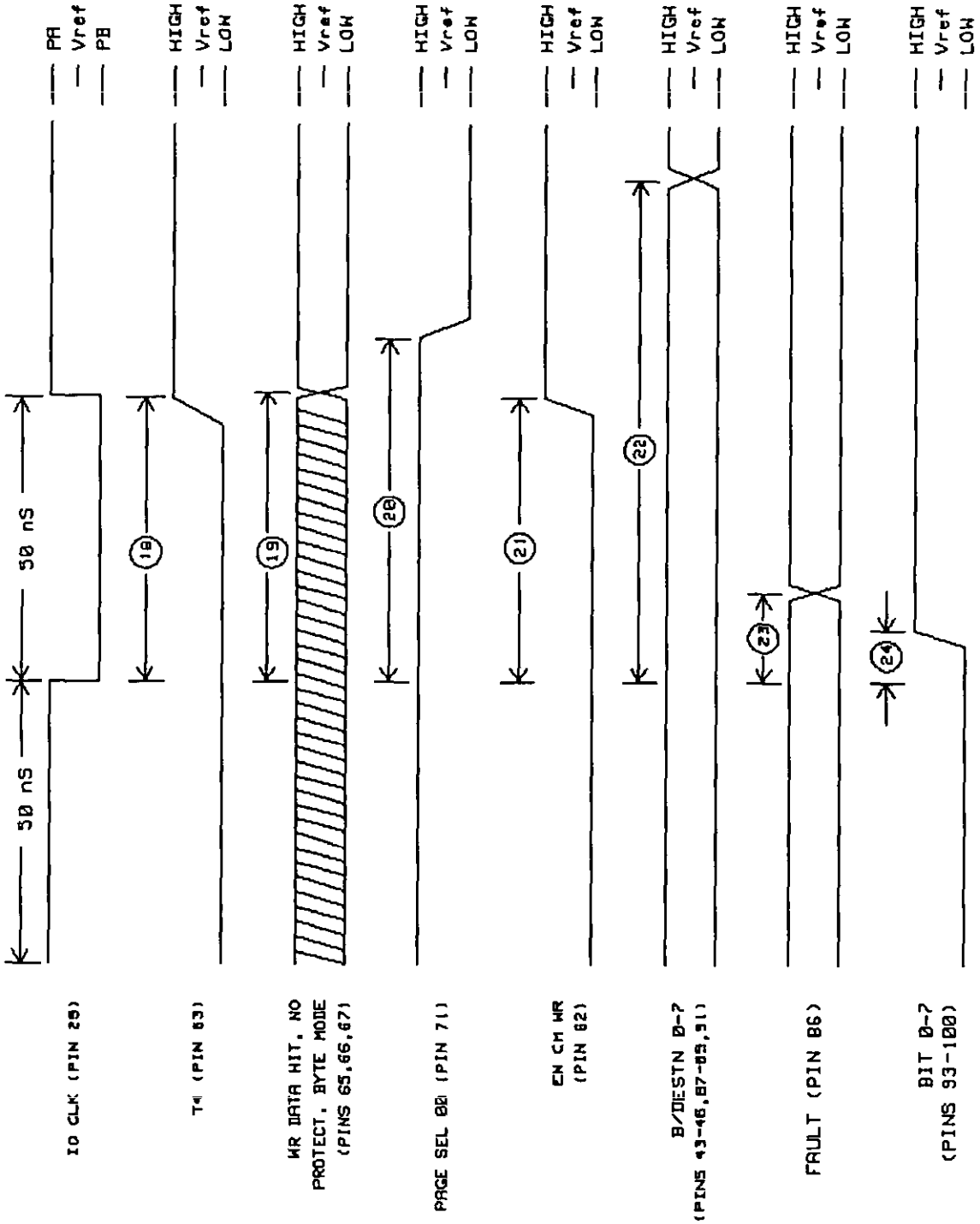


FIGURE 12d. Dynamic waveforms for table XVI.

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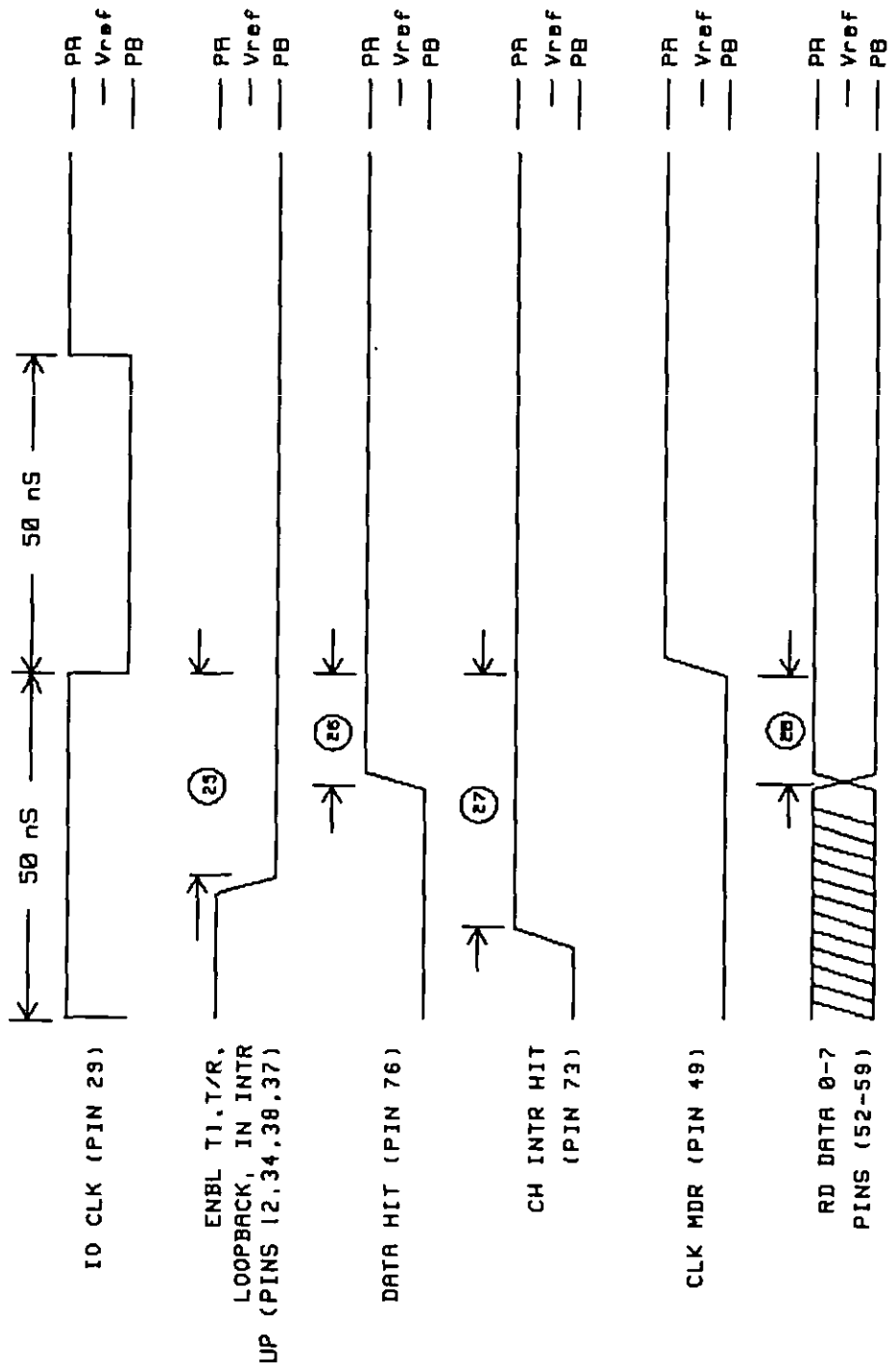


FIGURE 12e. Dynamic waveforms for table XVI.

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TABLE XVII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	EN INTR DATA	2	4	75	pF	1-4
R2	ADRS 0	3	2	50	pF	1-4
R3	ADRS 1	4	2	50	pF	1-4
R4	CI	5	4	75	pF	1-4
R5	CII	6	4	100	pF	1-4
R6	CIII	7	6	125	pF	1-4
R7	ENBL T1	12	2	75	pF	1-4
R8	MACLR	13				
R9	CII EN	16				
R10	CI EN	17				
R11	LD CII STK	19				
R12	LD CI STK	20				
R13	COMM PNDNG	21				
R14	MAE IN	24				
R15	EN CIII INTR	26	2	75	pF	1-4
R16	T SCAN	27	2	50	pF	1-4
R17	IO CLK	29	20	211	pF	1-4
R18	T/R	34	6	100	pF	1-4
R19	MPC MODE	36	16	225	pF	1-4
R20	IN INTR UP	37	4	125	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	LOOPBACK	38	2	50	pF	1-4
R22	GRANTED	41				
R23	ABORT	42				
R24	ADRS TO	47				
R25	DATA TO	48	2	50	pF	1-4
R26	CLK MDR	49	4	125	pF	1-4
R27	PARITY	50	2	50	pF	1-4
R28	S/D LTC	72	2	50	pF	1-4
R29	CH INTR HIT	73	6	100	pF	1-4
R30	LD CTRL WD	74	2	50	pF	1-4
R31	DATA HIT	76	6	100	pF	1-4
R32	CHAN 1	79	2	50	pF	1-4
R33	CHAN 0	80				
R34	CHAN 2	82				
R35	CHAN 3	83	2	50	pF	1-4
R36	MDR + T3	84	4	100	pF	1-4
R37	IR BIT 8	85	4	75	pF	1-4
R38	BUFF COMP	92	2	75	pF	1-4
Bidirectionals						
R39	B/DESTN 3	43	2	75	pF	1-4
R40	B/DESTN 2	44	2	75	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	B/DESTN 1	45	2	75	pF	1-4
R42	B/DESTN 0	46	2	75	pF	1-4
R43	RD DATA 07	52	4	75	pF	1-4
R44	RD DATA 06	53				
R45	RD DATA 05	54				
R46	RD DATA 04	55				
R47	RD DATA 03	56				
R48	RD DATA 02	57				
R49	RD DATA 01	58				
R50	RD DATA 00	59	4	75	pF	1-4
R51	B/DESTN 7	87	6	100	pF	1-4
R52	B/DESTN 6	88				
R53	B/DESTN 5	89	6	100	pF	1-4
R54	B/DESTN 4	91	6	75	pF	1-4
R55	BIT 7	93	4	75	pF	1-4
R56	BIT 6	94	4	75	pF	1-4
R57	BIT 5	95	6	100	pF	1-4
R58	BIT 4	96	6	125	pF	1-4
R59	BIT 3	97	4	75	pF	1-4

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TABLE XVII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R60	BIT 2	98	6	100	pF	1-4
R61	BIT 1	99	6	100	pF	1-4
R62	BIT 0	100	4	75	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R62
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R62
Capacitance measurement	NA	± 3.55 pF	R1
	NA	± 2.4 pF	R2,R3
	NA	± 3.55 pF	R4
	NA	± 4.8 pF	R5
	NA	± 5.95 pF	R6
	NA	± 3.65 pF	R7-R15
	NA	± 2.4 pF	R16
	NA	± 9.55 pF	R17
	NA	± 4.7 pF	R18
	NA	± 10.45 pF	R19
	NA	± 6.05 pF	R20
	NA	± 2.4 pF	R21-R25
	NA	± 6.05 pF	R26
	NA	± 2.4 pF	R27,R28
	NA	± 4.7 pF	R29
	NA	± 2.4 pF	R30

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 4.7 pF	R31
	NA	± 2.4 pF	R32-R35
	NA	± 4.8 pF	R36
	NA	± 3.55 pF	R37
	NA	± 3.65 pF	R38-R42
	NA	± 3.55 pF	R43-R50
	NA	± 4.7 pF	R51-R53
	NA	± 3.45 pF	R54
	NA	± 3.55 pF	R55,R56
	NA	± 4.7 pF	R57
	NA	± 5.95 pF	R58
	NA	± 3.55 pF	R59
	NA	± 4.7 pF	R60,R61
	NA	± 3.55 pF	R62

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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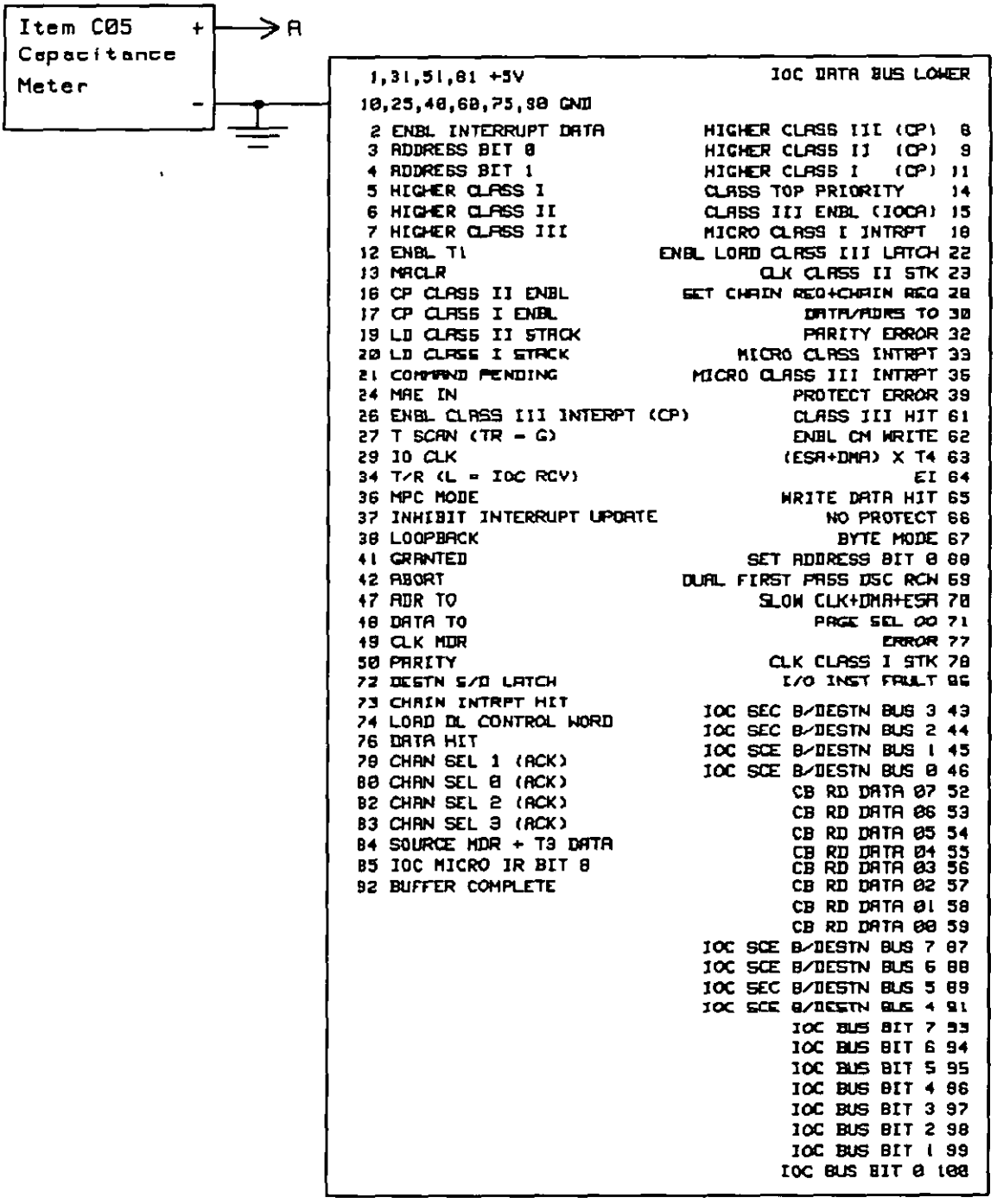


FIGURE 13. Input and output capacitance test circuit for table XVII.

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TABLE XVIII. Input clamp diode latch up test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
S	IO CLK	29	-18.0	-1.2	2.4	V	1-3

NOTES:

1. Input clamp diode latch up test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	S
Power supply current limit	1.49 A	NA	S
Forcing current, (Iin)	-18 mA	± 1.8 mA	S
Pattern voltage high, (VH) low, (VL)	NA NA	NA NA	S S
Input voltage measurement	NA	± 180 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XVIII and the voltage then measured.

(d) Apply pulse to the module for a duration of 500 ± 20 nS.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.49 A	NA	T1,T2
Pattern voltage high, (VH)	3.0 V	± 100 mV	T1,T2
low, (VL)	0.0 V	± 100 mV	T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

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NOTES: - Continued.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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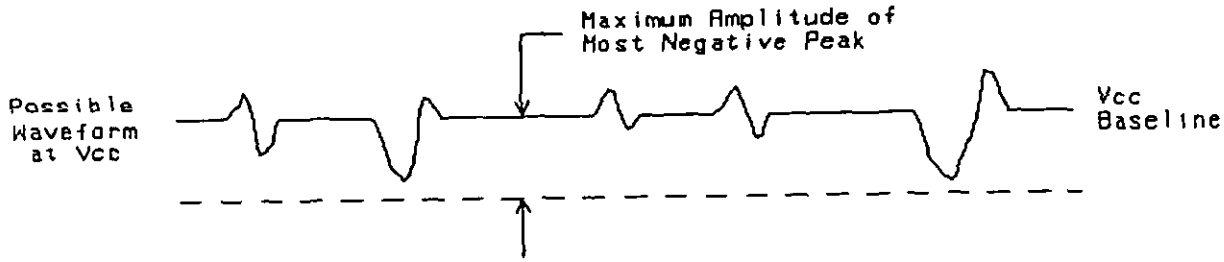
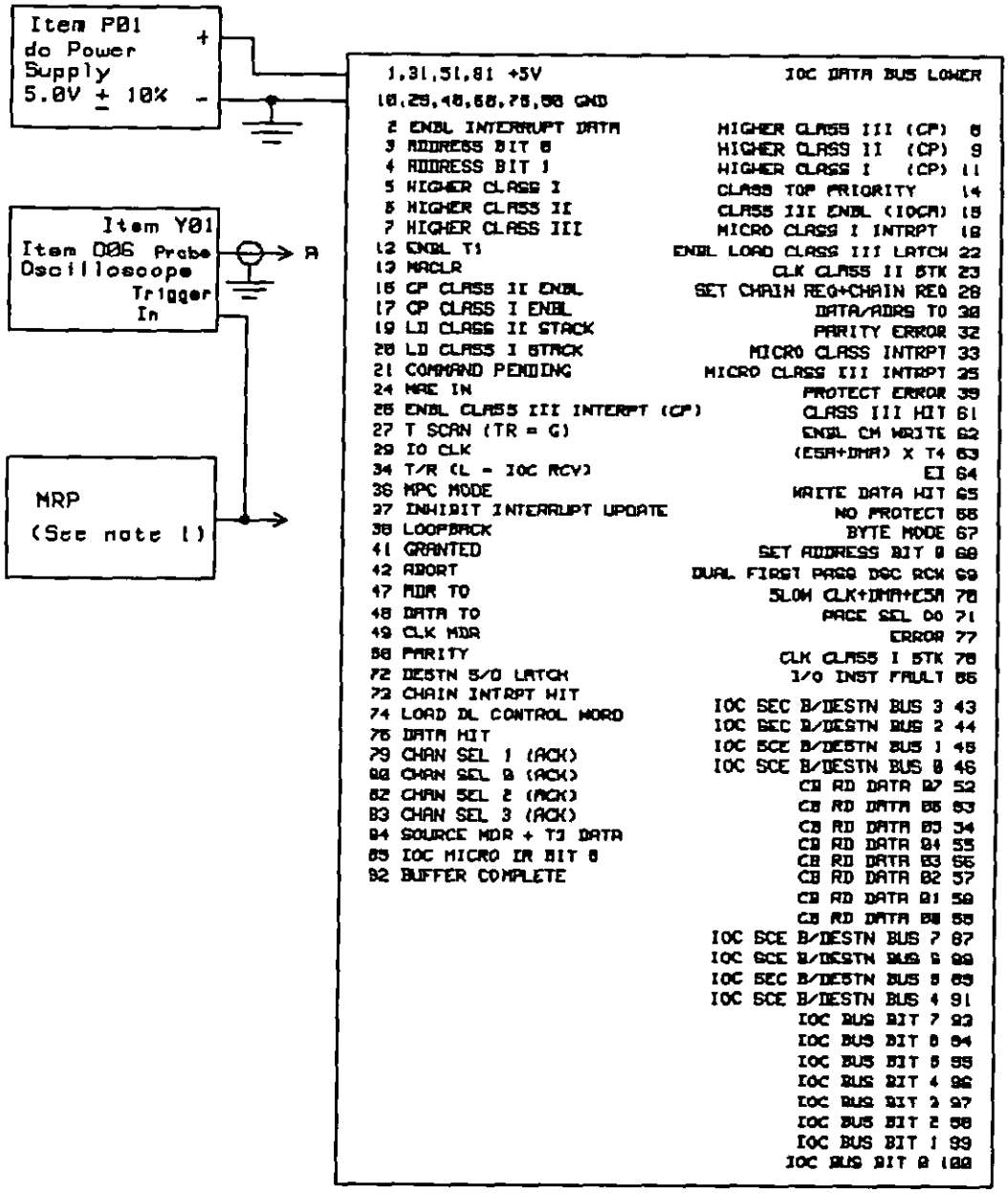


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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